



Diabetic Retinopathy Super Resolution & Zero-Shot Learning Benchmark

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This thesis closes a journey that was far more than research. It has been years of learning how to build, how to think, and how to continue when the weight of the work and life outside it felt overwhelming. There were moments when progress came easily, and others where it felt as if nothing would move no matter how hard I tried. Finishing this thesis is not simply the end of a project. It is the quiet proof that persistence endures even when certainty does not.

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1. Abstract

Diabetic retinopathy is a leading cause of preventable blindness, yet the earliest stages remain difficult to recognise because clinically relevant abnormalities are faint and often suppressed by real imaging degradation. Supervised deep learning systems achieve strong performance on curated datasets, but their reliability decreases when images originate from different clinics or contain subtle lesion patterns. Vision–language models and zero-shot learning provide an alternative that does not require large annotated datasets, but their effectiveness ultimately depends on the visibility of the pathology described by the prompt. This thesis examines whether enhancing fundus image visibility using super-resolution can improve zero-shot diabetic retinopathy screening under realistic acquisition variability.

Two publicly available datasets, APTOS and IDRiD, were used to evaluate the link between lesion visibility and prompt-based classification. A controlled degradation pipeline simulated screening-style variability in illumination, blur and resolution. Four enhancement methods were evaluated: bicubic interpolation, SRCNN, FSRCNN and ESPCN. Visibility improvements were quantified using image-level metrics and lesion-specific scoring, and zero-shot classification was assessed using CLIP and RET-CLIP across binary and five-class configurations.

The findings show that super-resolution improves the visibility of microaneurysms, shallow haemorrhages and other fine-scale abnormalities, leading to measurable gains in zero-shot classification. The effect is strongest in the earliest disease stages and persists across both datasets, indicating that improvements are not linked to dataset-specific imaging patterns. The performance advantage decreases in moderate and severe stages where lesions are already conspicuous. These results demonstrate that increasing visual fidelity can meaningfully enhance the reliability of zero-shot retinal disease screening and present a pathway for scalable early detection without the need for additional annotation resources.

2. Introduction

2.1 Clinical Background of Diabetic Retinopathy

Diabetic Retinopathy (DR) is a progressive microvascular complication of diabetes and the leading cause of preventable blindness among working-age adults worldwide [1]. More than one third of individuals with diabetes develop DR, and the prevalence is expected to rise in parallel with increasing global diabetes rates [2]. In the early stages, DR advances silently as microvascular damage accumulates in the retina without noticeable symptoms, which results in many patients presenting only after permanent vision loss has already occurred [3]. Clinically, the disease spans a continuum from microaneurysms and haemorrhages to vision-threatening proliferative complications such as neovascularisation, macular oedema and retinal detachment [4]. Although early DR is highly treatable, late-stage intervention rarely restores full visual function and imposes a significant socioeconomic burden on healthcare systems and affected individuals.

Early intervention depends on regular screening, yet screening attendance remains low due to reliance on manual retinal assessment, specialist shortages and limited accessibility in underserved regions [2]. Automated fundus image analysis is therefore viewed as a pathway to expanding screening capacity at population scale [4]. However, clinical deployment requires classification systems that perform reliably across real-world imaging variability. Fundus images collected in routine practice often contain blur, noise, illumination variation and low resolution, which can obscure early vascular abnormalities and lead to missed diagnoses [3]. Addressing the loss of diagnostic visibility under degraded imaging conditions forms the medical motivation for the present study.

2.2 Challenges in Early DR Detection and Fundus Imaging

Detecting the earliest signs of Diabetic Retinopathy (DR) remains one of the most demanding tasks in retinal screening because the first abnormalities often appear faint and nearly indistinguishable from healthy retinal tissue. Microaneurysms, tiny haemorrhages and small lipid deposits frequently resemble background vasculature or illumination artefacts, meaning that pathology can exist even when the eye appears clinically normal upon first inspection. Ophthalmologists routinely report that early DR requires magnification and controlled imaging

conditions to recognise confidently [1, 2]. The difficulty of recognising pathology in a seemingly healthy retina is illustrated in Figure 2.1. Although the left image appears normal, the lesion overlay on the right highlights subtle abnormalities that are extremely easy to miss during routine screening.

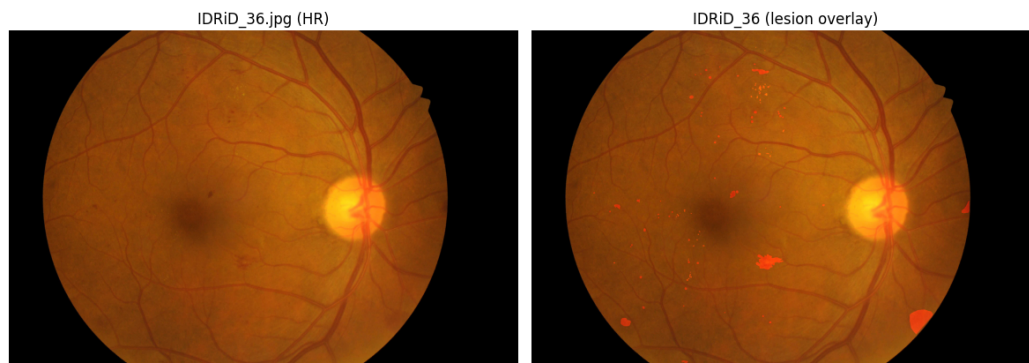


Figure 2.1: IDRiD fundus image with corresponding lesion overlay [5].

A second factor that complicates early DR detection is the variation in fundus image appearance across real clinical settings. Camera manufacturers implement different optical components, illumination systems and colour calibration procedures, meaning that the same pathology can appear noticeably different depending on the device used [2]. Patient-related factors such as limited pupil dilation, fixation instability and tear-film reflection introduce additional variability during acquisition, while clinic-level workflow conditions influence the degree of blur and brightness in captured images [4]. Even small fluctuations in image clarity can conceal microaneurysms and soft exudates, causing early pathology to resemble normal vasculature. These inconsistencies lead to frequent disagreement between graders and contribute to missed diagnoses during the earliest disease stages [1, 6]. Figure 2.2 illustrates this challenge by showing how early lesions become more visually apparent only as the disease progresses.

These complications are amplified in large-scale screening programs where image acquisition takes place across different clinics, populations and operators. Current public datasets capture only a fraction of this variability, which limits their ability to reflect real clinical diversity [7]. Studies evaluating cross-centre generalisation repeatedly show that models trained on one dataset experience reduced performance when tested on images from different sources, even when disease severity is identical [8]. This effect becomes even more pronounced for early DR, where the visibility of lesions lies very close to the threshold of detection and small differences in appearance directly influence diagnostic confidence. Recent multicentre evaluations further demonstrate that grader disagreement is highest during the earliest two disease stages because the same microaneurysms, exudates or faint haemorrhages can appear noticeably different depending on image sharpness, illumination and sensor calibration [6].

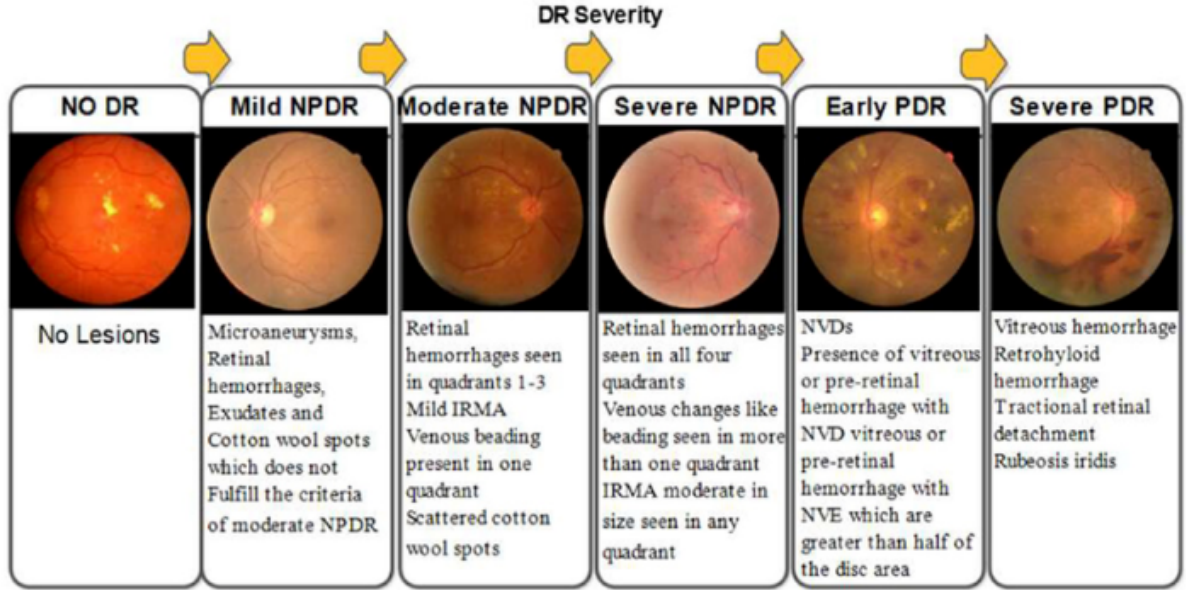


Figure 2.2: Clinical progression of DR severity and lesion visibility [9].

Annotation introduces an additional bottleneck for screening system development. Pixel-level lesion annotations require trained graders and are expensive and time-consuming to produce at scale. As a result, publicly available retinal datasets differ greatly in annotation granularity and class distribution [7, 10]. Since early DR occurs far less frequently than healthy cases, supervised models often focus on dominant visual patterns rather than the subtle indicators that matter clinically. Collectively, these factors explain why early-stage DR continues to be one of the most difficult retinal conditions to detect consistently in routine screening environments.

2.3 Deep Learning for DR Strengths and Limitations

Deep learning has transformed automated diabetic retinopathy (DR) screening through convolutional and transformer-based architectures trained on large labelled retinal datasets. These models achieve high sensitivity and specificity on curated benchmarks where lesion visibility is clear and training and testing distributions are well aligned [4, 6]. Public datasets such as EyePACS, DDR, Messidor and APTOS have supported this progress and enabled state-of-the-art staging performance as models evolved from CNNs to attention-based architectures [11, 12].

However, strong reported performance is closely tied to annotation scale and dataset consistency. Early DR cases are less common than healthy or advanced stages, which introduces class imbalance, and lesion-level annotation is costly to obtain [10]. Performance also decreases when models are tested on unseen imaging centres or different acquisition hardware, where changes in blur, colour calibration and illumination alter the appearance of subtle lesions [7,

8]. These trends suggest that supervised architectures excel under ideal image conditions but struggle with early DR and cross-centre variability. To contextualise these outcomes across the literature, Table 2.1 summarises representative works and the most frequent strengths and limitations reported.

Study	Model Type	Dataset(s)	Key Outcome and Limitation
Sahlsten et al. (2019) [4]	CNN	Kaggle EyePACS	High grading accuracy; performance reduced when image appearance varies across cameras.
Li et al. (2022) [6]	CNN Ensemble	Multicentre dataset	Strong DR and DMO results; lower agreement across imaging centres.
Men et al. (2023) [11]	Transformer	DDR, EyePACS, Messidor	State-of-the-art staging performance; requires large labelled datasets.
Kobat et al. (2022) [13]	DenseNet and patching	DIARETDB1	High lesion sensitivity; patch inference is computationally expensive.
Sebastian et al. (2023) [1]	Survey (multiple models)	Multiple benchmarks	Comprehensive comparison across architectures; highlights dependence on annotated data.

Table 2.1: Representative deep learning studies for DR detection and staging.

Taken together, these studies show that while deep learning models perform well under ideal conditions, reliability decreases when early DR lesions are faint or when imaging characteristics differ from the training distribution. This provides motivation for approaches that improve lesion visibility or reduce reliance on labelled datasets when operating under real clinical variability.

These findings reinforce that strong performance on curated datasets does not automatically translate to consistent screening outcomes across clinical environments. Models are most confident when lesions are clearly visible and when training and testing images share similar visual characteristics, but reliability decreases when confronted with variability in acquisition hardware, patient-level factors or image quality [8]. The dependency on large annotated datasets further limits scalability in regions where annotation resources are scarce. Consequently, despite major progress, supervised deep learning alone has not fully addressed the need for early DR detection under real clinical conditions, motivating the exploration of label-efficient alternatives such as zero-shot learning and multimodal vision-language models.

2.4 Zero-Shot Learning and Vision-Language Models for Medical Imaging

Zero-shot learning (ZSL) and vision-language models (VLMs) provide an alternative to conventional supervised learning for diabetic retinopathy (DR) screening. Rather than relying on large labelled datasets, these models align visual and textual features in a shared embedding space. The original CLIP framework showed that an image can be associated with a natural language description and classified without task-specific training [14]. MedCLIP adapts this approach to clinical imaging by incorporating medical terminology and specialist datasets, resulting in stronger alignment between fundus photographs and disease descriptions [15]. The advantage of this approach lies in the ability to recognise a condition from a descriptive prompt instead of a fixed label set. This is highly relevant for DR, where collecting lesion-level annotations for every disease stage and every possible phenotype is neither scalable nor sustainable. In principle, ZSL enables a model to interpret pathology even if that particular stage or description was not explicitly present during training, which makes it an attractive candidate for large population-based screening.

Recent VLM studies demonstrate that prompt-driven classification can detect a wide range of ocular and systemic abnormalities from retinal photographs [16]. In parallel, generalised zero-shot learning (GZSL) enables simultaneous recognition of both previously seen and unseen disease categories under the same inference setting [17]. These advances shift the focus from dataset expansion toward improved clinical expressiveness, where diagnostic prompts can be modified to refine or narrow the description of pathology. Taken together, these developments position ZSL as a practical means of extending DR screening capacity without continuously expanding annotation requirements.

Study	Framework Type	Task	Strength	Limitation	Observed
Wang et al. (2022) [15]	MedCLIP (VLM)	Medical classification	Strong text-image alignment	Difficulty with subtle lesion distinction	
Bravo-Rocca et al. (2025) [18]	Zero-shot clustering	DR identification	Enables continual learning	Still reliant on clear structural abnormalities	
Mahapatra et al. (2021) [17]	GZSL	Medical image classification	Recognises seen and unseen classes	Sensitive to dataset shift between imaging centres	
Su et al. (2024) [19]	RURA-Net (ZSL)	Disease diagnosis	Removes need for labelled training data	Performance loss on low-quality or degraded images	
Liu et al. (2024) [20]	VPL (Proxy learning)	Zero-shot diagnosis	Efficient inference and training	Limited lesion-level reasoning	
Aghabeigi Alooghareh et al. (2025) [16]	VLM	Ocular and systemic detection from fundus	Multi-condition detection	Requires disease to cause visible structural change	

Table 2.2: Representative zero-shot and vision-language approaches applied to medical imaging and retinal analysis.

Although VLMs decrease dependence on annotated datasets, their effectiveness for DR depends on the visibility of early lesions. Microaneurysms, faint haemorrhages and small exudates are often difficult to separate from normal retinal anatomy. Multiple studies report a consistent decline in classification accuracy when these early abnormalities are present because they lie close to the resolution and contrast limits of standard fundus imaging [10, 21]. Image degradation further compounds this effect. Blur, noise or illumination variations that are common in real clinical acquisition can suppress visual cues to the point where VLMs cannot separate early abnormalities from healthy tissue [22].

A further limitation is dataset shift across imaging centres. VLMs trained on one fundus dataset do not consistently generalise to images from other clinics or devices even when disease severity is the same [6, 23]. Prompt engineering can reduce this gap, but performance improvements are limited for early DR because the defining features are subtle and easily obscured. Current literature consistently shows the strongest VLM performance in moderate and severe stages of DR where lesions are large and structurally distinct [4].

Overall, ZSL and VLMs present a viable direction for large-scale DR screening with minimal annotation requirements. However, their effectiveness remains dependent on lesion visibility and on consistent image acquisition quality. These observations indicate a need for methods that improve the clarity of early lesions before VLM inference.

2.5 Super-Resolution for Enhancing Retinal Lesion Visibility

Super-resolution (SR) techniques aim to improve the spatial detail of an image beyond its native acquisition resolution. In retinal screening this is clinically meaningful because the earliest signs of diabetic retinopathy (DR) are small and faint, lying close to the visibility threshold of standard fundus imaging. Microaneurysms, soft exudates and shallow haemorrhages occupy only a few pixels in low-resolution fundus photographs, meaning their texture and borders can be lost when acquisition quality is reduced. SR methods therefore seek to recover vessel sharpness, lesion edges and local contrast so that subtle abnormalities become easier to identify in both manual assessment and computational pipelines [24, 25].

Study	Model Type	Enhancement Focus	Strength	Limitation Observed
Jia et al. (2024) [25]	GAN with vascular prior	Vessel clarity and lesion borders	Improves fine anatomical detail	Requires vascular masks for full gain
Lin et al. (2025) [24]	CNN / Review of SR models	Global sharpness and contrast	Enhances feature visibility for classifiers	Less effective with severe illumination variation
Hossain et al. (2023) [26]	Swin-FSR (Transformer)	Local texture refinement	Preserves shape while increasing sharpness	Very high computational cost
Aghabeigi Alooghareh et al. (2025) [16]	SR enhancement module	Disease-cue visibility for multi-condition detection	Improves clarity across varied patient groups	Limited benefit on high-quality clinical datasets

Table 2.3: Representative super-resolution methods evaluated for retinal imaging enhancement.

SR becomes particularly relevant when retinal screening occurs across multiple clinics or imaging devices. Differences in illumination, optical configuration and sensor resolution produce variations in sharpness and colour balance that can suppress early abnormalities. Several studies report performance reductions in DR classifiers when high-resolution training data are evaluated on lower-quality community screening images [8]. SR offers a practical way to reduce this decline by restoring fine vascular structure and lesion borders before classification. Evidence from manual grading studies also indicates that clinicians report higher confidence and sensitivity when high-resolution retinal scans are available [10].

The developments across supervised learning, zero-shot classification and super-resolution show that each direction addresses a different aspect of the early DR screening problem. However, these directions are progressing largely independently, and the interaction between lesion visibility enhancement and prompt-based classification has not yet been tested. This raises an important question: if visibility is the limiting factor for early DR detection, can improving visibility unlock the full potential of ZSL-based screening?

Overall, SR has the potential to reduce missed early abnormalities by recovering clinically important texture and local contrast. Its utility within zero-shot learning remains largely unexplored, creating a clear motivation to evaluate whether enhancing lesion visibility prior to ZSL inference can improve early DR detection.

2.6 Research Gap and Contribution

Research on diabetic retinopathy (DR) spans three main directions: supervised deep learning classifiers, super-resolution (SR) methods for retinal enhancement, and zero-shot or vision-language approaches for label-efficient screening. Each has progressed independently, yet none addresses the combined challenge of recognising early DR when abnormalities are faint and image quality varies across clinical acquisition settings.

Supervised convolutional and transformer-based DR classifiers achieve high accuracy on controlled datasets, but their performance decreases when images originate from unseen centres or contain subtle early lesions [2, 8]. SR techniques enhance lesion visibility and vascular clarity, and several studies report improvements for manual grading and supervised pipelines [24, 25]. However, these works do not examine whether SR improves performance for prompt-driven screening. Zero-shot learning (ZSL) and vision-language models (VLMs) bypass the need for large annotated datasets by aligning retinal images with natural-language disease descriptions [15, 16]. Yet recent findings show that their accuracy declines when abnormalities are small, low-contrast or affected by blur and illumination variation [21].

Across the literature, no prior study evaluates whether improving lesion visibility through SR can increase the diagnostic reliability of ZSL for early DR, nor whether this holds under realistic variation in image quality. This gap matters because the success of ZSL depends on the recognisability of the abnormalities described by text prompts. Now, to the best of current literature, no prior study has evaluated whether enhancing lesion visibility through SR can improve ZSL-based DR classification under clinically realistic variations in acquisition settings. Therefore, this work evaluates whether improved visibility alone can strengthen ZSL-based screening, offering a pathway where performance gains arise from image enhancement rather than costly annotation expansion.

Research Area	Current Status and Limitations
Supervised CNN and ViT models for DR classification	High accuracy on curated datasets, but reduced reliability for early DR and under cross-centre variation.
Super-resolution for retinal image analysis	Improves visibility of fine vascular and lesion structures; not previously evaluated for zero-shot or prompt-based DR classification.
Zero-shot learning and VLM-based DR screening	Reduces reliance on labels; performance declines when lesions are subtle or when image quality varies.
Intersection of SR and ZSL for early DR (this thesis)	Impact of lesion visibility enhancement on ZSL classification under realistic image variability remains untested.

Table 2.4: Summary of related research areas and unresolved gaps addressed in this thesis.

This thesis contributes new evidence at the intersection of SR and ZSL for DR screening. It introduces, to the best of current literature, the first systematic evaluation of whether enhancing lesion visibility through SR improves prompt-driven DR classification. The study (i) quantifies lesion visibility gains produced by SR using both image-level and lesion-specific metrics, and (ii) measures whether these gains translate into improved ZSL classification across multiple datasets and clinically realistic variations in illumination, blur and resolution. By isolating the effect of visibility enhancement on text-prompt inference, this work demonstrates whether ZSL can benefit from image enhancement rather than additional annotation, providing insight into a screening pathway that is both scalable and clinically practical.

2.7 Research Purpose, Research Questions, and Hypotheses

Early diabetic retinopathy (DR) remains the most clinically significant stage for preventing vision loss, yet it is also the most difficult to detect reliably in population screening. Zero-shot learning (ZSL) and vision-language models (VLMs) present a promising alternative to conventional supervised systems because they do not require lesion-level annotations for every disease stage. However, their performance is ultimately constrained by the visibility of the abnormalities described by natural-language prompts. When lesions are small, low-contrast or affected by acquisition artefacts, ZSL models struggle to associate the visual evidence with the correct textual description, even when the prompt is clinically precise. This suggests that ZSL failure in early DR does not arise from semantic limitations, but rather from insufficient lesion visibility in the visual input.

The purpose of this research is therefore to determine whether enhancing fundus images with super-resolution (SR) improves the detectability of early DR for ZSL-based classification under realistic clinical variability. The study evaluates both (i) the effect of SR on the visual clarity of subtle lesions and (ii) the extent to which these visibility gains translate into improved ZSL performance across datasets captured with different imaging devices and acquisition protocols. The overarching goal is to assess whether early DR screening can be strengthened by improving the image itself rather than by increasing the volume of annotated data. To operationalise this aim, the thesis addresses the following research questions:

- **RQ1:** Does super-resolution enhance the visibility of fine-grained DR features such as microaneurysms, haemorrhages and soft exudates in early disease stages?
- **RQ2:** Does improved lesion visibility lead to measurable gains in ZSL classification performance for early DR when natural-language prompts are used?
- **RQ3:** Do potential ZSL improvements generalise across datasets that differ in imaging devices, illumination and acquisition conditions?
- **RQ4:** Is the effect of super-resolution greatest for early DR compared with moderate or severe stages where lesions are already visually prominent?

These questions correspond to the following hypotheses:

- **H1:** Super-resolution produces measurable improvements in the visibility of early DR lesions based on objective image-quality metrics and qualitative expert assessment.
- **H2:** Increased lesion visibility results in higher ZSL classification performance for early DR compared with native-resolution images.

- **H3:** ZSL performance gains due to SR persist across datasets with varied imaging hardware and acquisition settings.
- **H4:** The benefit of SR is most pronounced for the earliest DR stages and declines as lesions become larger and visually conspicuous.

Together, these hypotheses establish a direct link between image quality, lesion visibility and the effectiveness of prompt-based retinal screening. If validated, the findings would demonstrate that improving the visual input to ZSL may be as impactful as increasing annotation volume, offering a practical and scalable pathway for early DR detection across diverse clinical environments.

3. Methodology

3.1 Datasets and Cohort Composition

3.1.1 Dataset sources, licensing and acquisition

This study employs two publicly available retinal fundus datasets widely used in diabetic retinopathy (DR) research: the Asia Pacific Tele-Ophthalmology Society Blindness Detection dataset (APTOS) [27] and the Indian Diabetic Retinopathy Image Dataset (IDRiD) [5]. The datasets were selected for their clinical relevance, high-quality severity annotations and contrasting imaging characteristics, which together provide a rigorous test-bed for evaluating model consistency under domain shift.

3.1.1.1 APTOS

The APTOS Blindness Detection dataset was obtained from the Kaggle competition platform under an educational licence and contains colour fundus photographs sourced from multiple ophthalmic clinics across Asia. Images were captured using heterogeneous retinal camera systems, resulting in pronounced variability in illumination, colour temperature, image resolution, field of view and focus quality. Unlike research-grade ophthalmic datasets, APTOS reflects the imaging diversity that arises in mass-screening environments where equipment, operator experience and patient cooperation cannot be standardised.

This heterogeneity makes APTOS a valuable benchmark for testing the resilience of automated DR detection systems. In particular, the large proportion of clinically healthy cases (class 0) and relatively sparse representation of advanced stages create a long-tailed distribution aligned with real-world screening deployments. Such class imbalance poses significant learning challenges, as early-stage DR (classes 1-2) often presents subtle textural changes that are easily overshadowed by the majority distribution. From a super-resolution (SR) and zero-shot learning (ZSL) perspective, APTOS acts as a stress test for feature reliability: if SR improves lesion visibility, and if ZSL effectively learns semantically transferable lesion cues, then performance gains should be most apparent under the noisy and imbalanced conditions characteristic of this dataset. All images were fully anonymised, and no additional ethical approval was required for academic use or ethical clearance

3.1.1.2 IDRiD

The Indian Diabetic Retinopathy Image Dataset (IDRiD) was downloaded from the IEEE DataPort repository under a research-only licence intended for non-commercial academic use. In contrast to APTOS, IDRiD images were acquired in a single clinical environment in India using a consistent imaging protocol, uniform illumination, and standardised retinal camera hardware. This controlled setting ensures high inter-image consistency and reduces the influence of acquisition artefacts, enabling clearer separation of disease severity based on retinal pathology rather than imaging conditions.

A distinguishing feature of IDRiD is the availability of both DR severity labels and pixel-level lesion annotations for exudates, microaneurysms, haemorrhages and neovascularisation. The dataset therefore provides not only global disease staging but also spatially grounded evidence for each classification. For interpretability and SR analysis, this is critical: enhanced visibility of small lesions should correlate with spatial annotations, and ZSL models should be capable of leveraging pathology-level cues that remain consistent across datasets even when global appearance shifts.

Furthermore, unlike the healthy-skewed APTOS distribution, IDRiD contains a higher concentration of moderate to severe DR cases. This creates a complementary evaluation setting in which the difficulty lies not in identifying subtle cases among many normals, but in reliably distinguishing multiple pathological stages with overlapping visual characteristics. Consequently, IDRiD enables evaluation of SR and ZSL models under clinically structured staging rather than population-screening imbalance. All images were anonymised at source and required no further ethical clearance.

3.1.2 Severity distributions and labelling methodology

Both datasets adopt the International Clinical Diabetic Retinopathy Severity (ICDR) Scale: no DR (0), mild (1), moderate (2), severe (3) and proliferative DR (4). Each level corresponds to recognisable clinical manifestations including microaneurysms, haemorrhages, cotton-wool spots, venous beading and neovascularisation. The clinical progression of DR severity used throughout this thesis is shown in Figure 3.1. This progression provides essential context for interpreting the class distributions and later stage-stratified model performance.

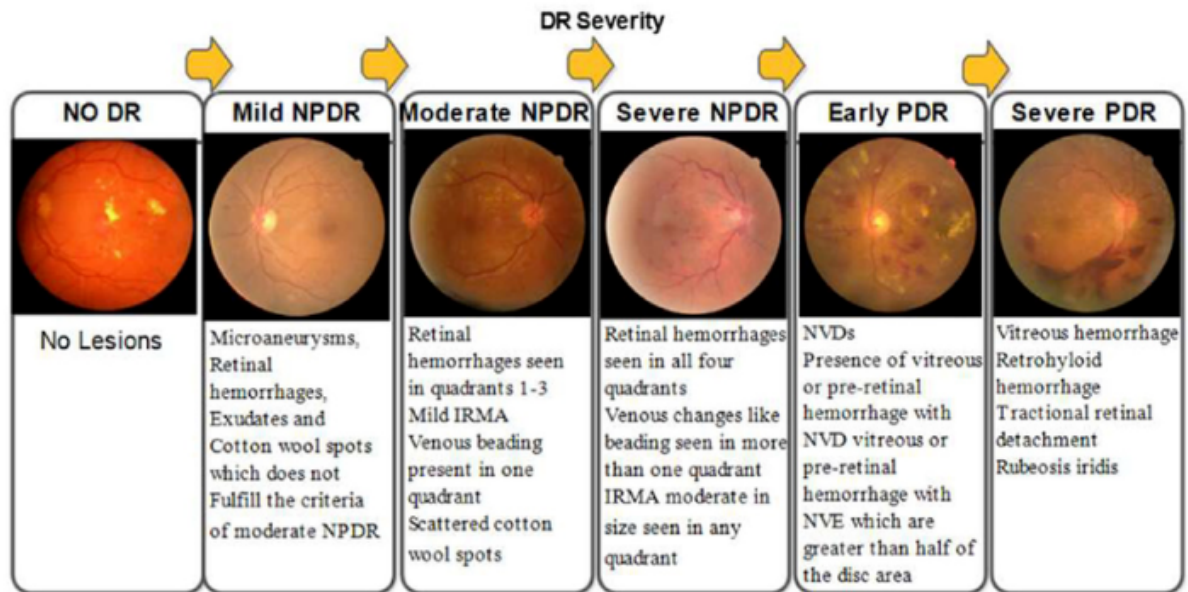


Figure 3.1: Clinical progression of DR severity and lesion visibility [9].

Class distributions were computed from the official training annotations for each dataset. APTOS displays a strong skew toward the no-DR category, which closely resembles the distribution encountered in routine screening programs. IDRiD contains fewer total samples but provides a more balanced representation across intermediate and advanced disease stages. These contrasting profiles are summarised in Table 3.1 and visualised in Figure 3.2a and Figure 3.2b.

Dataset	Most prevalent grade	Interpretation
APTOS	0 (No DR)	Real-world screening bias with many clinically normal retinas
IDRiD	2 (Moderate DR)	Greater density of pathological cases suitable for staging analysis

Table 3.1: Summary of severity distribution characteristics across datasets.

The contrasting class distributions have direct methodological implications for this study. APTOS reflects a population-level screening scenario in which most patients do not exhibit clinical signs of DR, meaning models trained on this dataset risk over-privileging dominant non-pathological visual patterns. In contrast, IDRiD represents a more diagnostically enriched cohort with clearer separation between disease stages, enabling models to learn pathological features relevant for clinical staging. Taken together, these differences provide complementary benchmarking conditions: APTOS tests a model’s adaptability under high class imbalance and heterogeneous image quality, whereas IDRiD evaluates its capacity to discriminate progressively subtle lesion patterns across the full severity spectrum. Consequently, improvements that generalise across both datasets are unlikely to be attributable to dataset-specific artefacts and instead reflect genuine advances in lesion visibility and feature extraction.

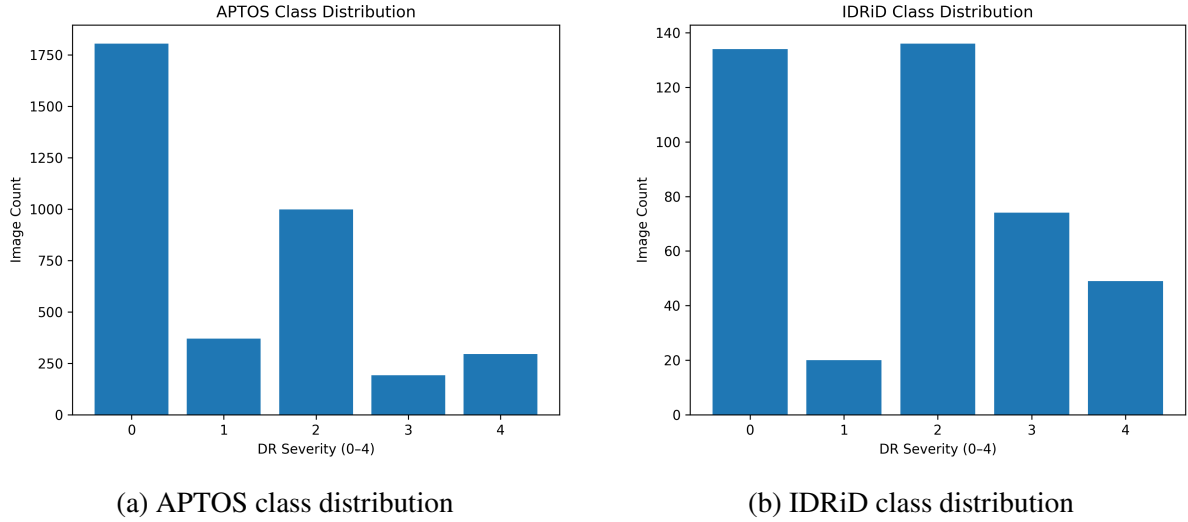


Figure 3.2: Comparison of severity prevalence between APTOS and IDRiD.

The distributions shown above have direct methodological implications. APTOS requires a model to distinguish subtle early abnormalities from a large pool of healthy cases, increasing the risk of missed early DR. IDRiD, by contrast, challenges the model to differentiate progressively worsening pathology across multiple adjacent stages. These complementary characteristics justify the combined use of both datasets in this thesis: APTOS stresses screening sensitivity, whereas IDRiD stresses staging discrimination. Later experimental sections return to these distributions when interpreting early-stage versus late-stage performance under super-resolution and zero-shot conditions.

3.1.3 Cross-dataset heterogeneity and clinical motivation

Although both datasets follow the ICDR grading framework, their imaging characteristics differ considerably. APTOS includes images collected from multiple clinics using different camera systems, which introduces variation in illumination, colour tone, contrast, focus and spatial resolution. IDRiD, in contrast, is captured within a single clinical environment under a consistent imaging protocol, resulting in stable lighting, field of view and uniformly high resolution. These differences are demonstrated in Figure 3.3, where APTOS shows a wide range of pixel dimensions and IDRiD forms a tight cluster around 4288×2848 pixels.

This variation has meaningful clinical implications. Screening programs in the real world are more likely to resemble APTOS, where technical artefacts and inconsistent imaging conditions frequently obscure early lesions such as microaneurysms and small haemorrhages. IDRiD, however, reflects a controlled diagnostic setting where fine disease structures are consistently visible and the challenge lies in distinguishing between degrees of pathology rather than com-

compensating for low-quality imaging. Evaluating models on both datasets therefore enables an assessment of their ability to operate across both noisy screening scenarios and high-quality clinical imaging.

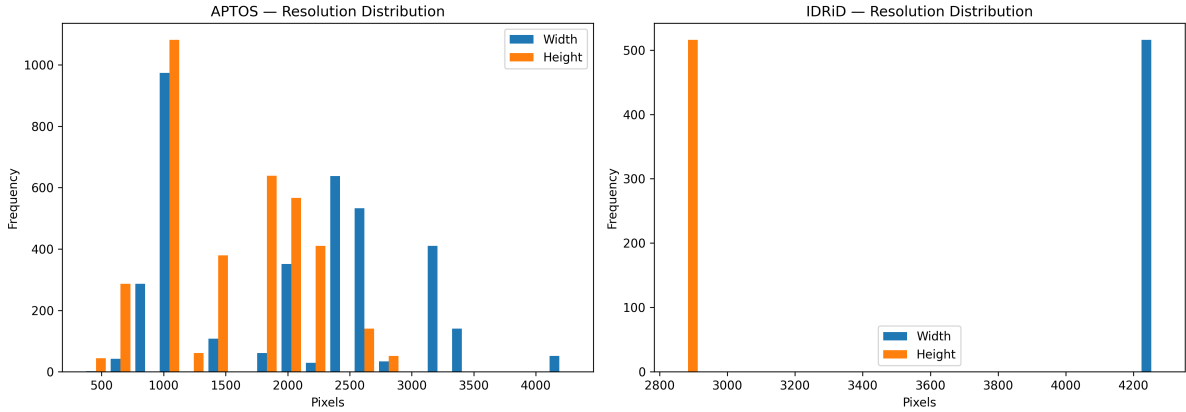


Figure 3.3: Resolution distribution comparison between APTOS and IDRiD. APTOS spans a wide range of capture dimensions, whereas IDRiD remains consistent due to a unified acquisition protocol.

3.1.4 Domain shift implications for SR and ZSL experiments

The acquisition differences between APTOS and IDRiD create a natural domain shift that is highly relevant to machine learning for diabetic retinopathy. Models trained on one dataset may not transfer seamlessly to the other if the visual cues they learn are tied to the dataset’s imaging characteristics rather than the underlying pathology. This issue is especially important for zero-shot learning (ZSL), where predictions depend on the alignment between visual lesion cues and language-based semantic descriptions.

Super-resolution (SR) provides a direct mechanism to address part of this shift by improving lesion visibility in lower-quality APTOS images. If SR has a meaningful effect, two outcomes are expected: (i) improvements in image visibility and ZSL performance on degraded or lower-resolution images, and (ii) reductions in the performance gap when models are transferred from one dataset to the other. If these improvements do not appear, then the limiting factor is likely the visual-text alignment process rather than visibility.

The differences between APTOS and IDRiD therefore offer a structured way to examine whether SR benefits ZSL in both screening-style and clinic-style imaging environments. Any gains that hold across both datasets indicate that the approach supports lesion-focused signal rather than dependence on dataset-specific imaging patterns.

3.2 Image Preprocessing and Degradation Simulation

3.2.1 Resolution normalisation and circular retinal cropping

The raw APTOS and IDRiD images vary considerably in spatial resolution due to differences in camera hardware and export settings. If the models were trained on native resolutions, anatomical structures such as the optic disc and macular region would appear at inconsistent relative sizes for reasons unrelated to disease. To avoid this effect, all images were resampled to a standard 512 x 512 resolution using bicubic interpolation. This provides a consistent effective receptive field for the encoder and prevents scale-driven bias during learning.

After resolution normalisation, a circular cropping step was applied to isolate the retinal region. A binary mask was produced using intensity thresholding, followed by morphological filtering to refine the retinal boundary. Pixels outside the retinal circumference were replaced with a black background. This removes peri-orbital tissue, camera borders and photographic noise while preserving all diagnostically important structures including the fovea, optic disc and vascular network.

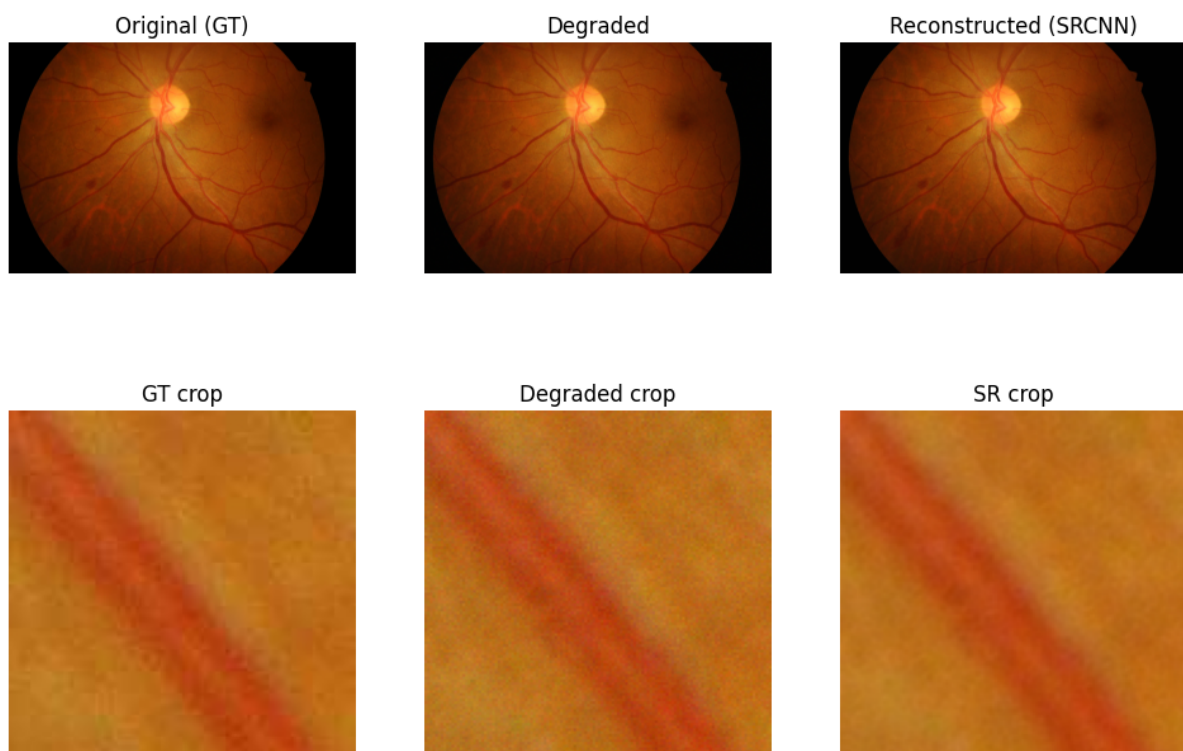


Figure 3.4: Illustration of the resolution normalisation and circular cropping output. Retinal anatomy is preserved while non-retinal context is removed.

The example in Figure 3.4 shows the effect of the cropping step. The field of view is aligned across patients regardless of camera brand, flash intensity or framing. This avoids the risk of the classifier inferring disease from irrelevant patterns such as eyelid visibility, camera shadowing or border reflections. All lesion morphology remains intact, which supports learning based on clinically meaningful structures.

3.2.2 Colour and illumination standardisation

Significant variation in colour temperature and lighting exists across fundus imaging systems, especially within APTOS where images originate from multiple clinics. Without correction, these differences can be memorised by models and treated as predictive features, reducing clinical validity. To mitigate this, colour correction was performed in CIE-Lab space which separates luminance and chromaticity.

The L channel was processed using adaptive histogram equalisation to stabilise brightness without amplifying noise. The a and b channels were shifted toward a reference distribution computed from a curated set of high-quality images. This approach reduces scanner-specific chromatic bias while preserving the natural pigmentation of retinal tissue.

In addition to improving dataset consistency, colour standardisation is necessary for the controlled evaluation of later experiments. Since the project examines whether super-resolution improves lesion visibility for zero-shot learning, the underlying colour characteristics must remain constant. If raw images retained their original illumination differences, it would become unclear whether improvements in classification originate from enhanced anatomical detail or from favourable colour bias linked to a particular scanner. Performing colour correction before applying super-resolution ensures that any change in visibility arises from structural refinement rather than unintended colour cues.

This step also addresses a practical limitation of automated DR screening noted in ophthalmic imaging literature. Algorithms often experience reduced sensitivity when evaluated on images with different colour profiles compared to those used during training. For example, strong flash illumination can wash out early microaneurysms, while warm sensor bias can hide subtle haemorrhages and venular changes. Standardising colour helps prevent these scanner-dependent variations from masking or accentuating pathology, increasing the likelihood that models learn clinically meaningful patterns.

A further advantage of the procedure is that global colour balance is corrected without altering local contrast. Although the palette of the image shifts toward a consistent distribution, the

pixel-level relationships that define pathology remain unchanged. Fine features such as vascular calibre, lesion edges and optic disc boundaries retain their original morphology. This is particularly important when combining APTOS and IDRiD, as it allows both datasets to act as a unified clinical population without discarding the inherent diversity of disease presentations.

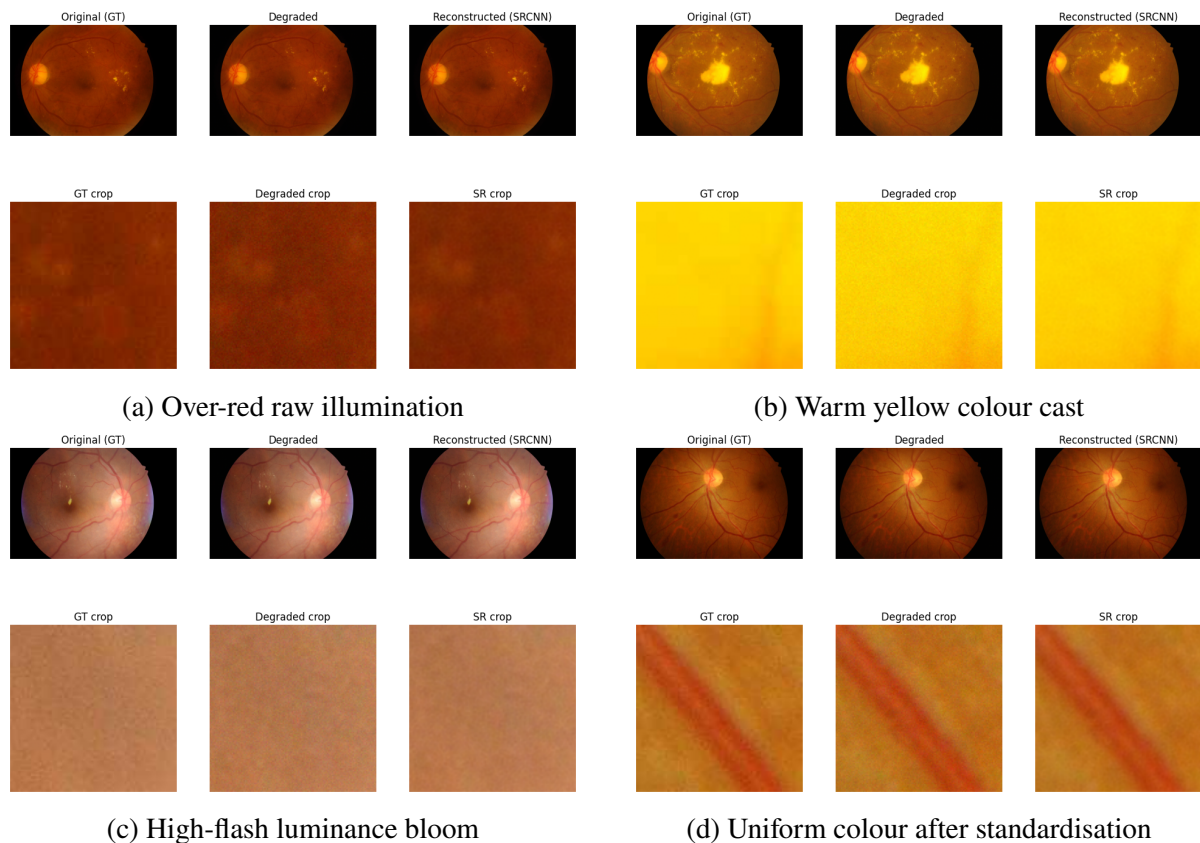


Figure 3.5: Examples of colour and illumination variability before standardisation (top row) and the corrected clinical appearance (bottom right).

This figure demonstrates the value of colour and illumination standardisation. The raw samples show large differences in hue and brightness despite similar clinical conditions. After correction, the retinal tissue displays a consistent appearance while maintaining lesion structure. Vessel calibre, microaneurysms and optic disc borders remain unchanged, confirming that the transformation adjusts global colour without modifying anatomy. This avoids models inferring disease likelihood from camera type or lighting and ensures that any visibility improvements reported later for super-resolution arise from the models rather than from preprocessing biases.

3.2.3 Artefact removal and quality filtering

The study retained all images provided in the dataset were retained without exclusion based on subjective quality criteria rather than excluding those with reduced visibility or problematic features as in practice this would not occur. In routine screening, fundus photographs are rarely discarded unless they are entirely unreadable, meaning that image quality varies widely across patients and across clinics [4]. Removing such samples would create an unrealistic experimental setting and risk inflating performance. Instead of filtering based on subjective quality criteria, the preprocessing stage focused on correcting only spatial framing and colour variation, as described in the previous subsections. This ensured that subsequent performance differences were driven by model capability rather than by incomplete dataset curation. The approach aligns with recommendations in recent surveys that emphasise evaluating diabetic retinopathy systems under realistic clinical image conditions [1, 8].

3.2.4 Clinically motivated degradation

Clinical acquisition variability is an established challenge in automated diabetic retinopathy assessment. Several studies report that loss of fine detail, uneven illumination and sensor noise contribute to reduced lesion visibility and misclassification [2, 6]. To model these patterns rather than idealised input, a controlled degradation pipeline was applied to every APTOS image. The procedure was designed to match realistic artefacts commonly encountered in community screening rather than to create synthetic damage.

The degradation consisted of sequential bicubic downsampling and upsampling to simulate low-resolution cameras, Gaussian blur with parameter σ to reflect soft focus and fixation error, and additive Gaussian noise to reproduce low flash intensity and sensor interference. The parameters were selected based on empirical inspection of clinical acquisition variability reported across multi-centre datasets [7, 10]. The degraded images serve as a visibility stress-test for the super-resolution models and allow later zero-shot learning experiments to examine failure cases that emerge when lesion contrast is weak but the retinal photograph still appears acceptable to clinicians.

3.2.5 Mathematical formulation of degradation process

For an original RGB fundus image I , the degraded image $\hat{I}$ was produced by applying down-sampling, upsampling, Gaussian blur and additive noise in the following order:

$$\hat{I} = \left(G_{\sigma} * (D_{\downarrow r} I) \uparrow r \right) + N(0, \sigma^2).$$

Here $D_{\downarrow r}$ denotes bicubic downsampling by factor r followed by bicubic upsampling to the original resolution, G_{σ} represents convolution with a Gaussian kernel of standard deviation σ , and $N(0, \sigma^2)$ is pixel-wise Gaussian noise with zero mean and variance σ^2 .

The formulation mirrors visibility degradation seen in fundus imaging systems [3, 22]. Down-sampling removes high-frequency structures such as vessel calibre variation and shallow haemorrhage boundaries, blur attenuates lesion margins, and noise affects chromatic uniformity. Because these mechanisms selectively weaken early-stage pathology cues, they provide a controlled basis for evaluating whether super-resolution improves lesion-level visibility rather than global brightness or contrast.

3.2.6 Manifest generation and directory structure

Each output image, including degraded and later super-resolved variants, was recorded in a manifest file for full traceability. For every sample, the manifest lists the original path, the derived output path, and the degradation parameters including scale factor r , blur parameter σ and noise standard deviation. This ensures that any downstream performance result can be linked back to the exact input image and transformation sequence.

The directory structure separates datasets (APTOS and IDRiD), imaging condition (original, degraded and super-resolved) and model type. Files maintain one-to-one correspondence by filename, allowing pairing of results across experiments without ambiguity. This structure supports controlled comparisons between visibility conditions and prevents accidental data duplication or omission. The manifest design follows recommendations for transparent benchmarking in medical imaging pipelines [24, 28].

3.2.7 Clinical degradation patterns and relevance to screening environments

Variability in retinal image quality is not random but follows identifiable patterns shaped by the constraints of real-world screening environments. Community-based diabetic eye screening programs, particularly in rural, low-resource and mobile-clinic settings, frequently operate with ageing optical hardware, inconsistent illumination and limited operator training [2, 7]. These factors are known to reduce the visibility of early-stage diabetic retinopathy even when the photograph remains diagnostically acceptable to a clinician.

Loss of high spatial detail commonly arises in facilities that rely on low-cost fundus cameras with small sensors or optical assemblies that have experienced mechanical wear. This produces the same attenuation of high-frequency vascular information that is simulated through bicubic downsampling and upsampling in this study. Gaussian blur reflects soft focus introduced by limited patient fixation stability during handheld camera operation, a challenge reported extensively in remote and outreach screening initiatives [16]. Additive noise maps to scenarios in which flash intensity is reduced to protect patient comfort or when power supply fluctuations affect sensor behaviour, conditions observed particularly in mobile and rural deployments.

The degradation pipeline therefore does not generate abstract distortions but models the visibility decline that repeatedly appears in low-resource screening environments. These impairments have been identified as key contributors to misclassification by automated diagnostic systems [1, 6]. By mirroring them in a controlled manner, the study enables later experiments to evaluate whether super-resolution meaningfully restores lesion visibility when image acquisition is constrained rather than ideal. This aligns with the objective of improving automated DR detection for the clinical conditions in which it is needed most, rather than only within laboratory imaging standards.

3.3 Super-Resolution Framework

3.3.1 Model selection and architectural justification

The SR component of this study was designed to enhance lesion visibility while preserving the clinical truthfulness of the fundus image. In diabetic retinopathy, diagnostic cues such as microaneurysms, haemorrhages and vascular distortions appear at very small spatial scales, and their presence must not be inferred or synthetically introduced. Research on fundus image enhancement emphasises that super-resolution models can reduce clinical reliability when they generate artificial high-frequency texture that resembles pathology [1, 10]. For this reason, the SR strategy in this study followed a conservative design philosophy, favouring architectures that recover detail that already exists rather than fabricating new structure.

Four SR methods were implemented to represent clinically safe enhancement behaviour: bicubic interpolation as a non-learnable baseline, and three lightweight convolutional networks (SRCNN, FSRCNN and ESPCN). These architectures were chosen because they satisfy three requirements that are important for real diabetic retinopathy screening:

- They do not rely on adversarial training objectives, which reduces the risk of generating artificial vessels or exudates;
- They exhibit predictable structural behaviour, meaning that the morphology of the output corresponds to the morphology of the input;
- They are computationally lightweight, which enables deployment in screening environments where GPU availability may be limited [8].

Recent studies have shown that lightweight CNN-based SR can improve the visibility of vascular and lesion structures without altering the topology of the retina, whereas adversarial and perceptual-loss SR models have been observed to produce spurious vascular loops and exudate-like artefacts that complicate clinical interpretation [24]. The purpose of SR in this study was therefore not to maximise sharpness or aesthetic appearance, but to stabilise visibility under degraded acquisition conditions while maintaining the fidelity of the original pathology.

This conservative design also improves interpretability of downstream results. If grading performance increases after conservative SR, the improvement can be attributed to genuine recovery of retinal structure. In contrast, if performance rises only under highly generative SR models, the enhancement may originate from hallucinated detail rather than clinical signal.

3.3.1.1 Bicubic baseline

Bicubic interpolation was included as a non-learnable reference to isolate the structural contribution of trainable SR models. Bicubic scaling enlarges an image by fitting a two-dimensional cubic polynomial over a 4×4 neighbourhood of pixels surrounding each output location [29]. Formally, for an output coordinate (x, y) , the interpolated value is computed as

$$I'(x, y) = \sum_{i=-1}^2 \sum_{j=-1}^2 w(i, j) I(x + i, y + j)$$

where I is the original image and $w(i, j)$ are spatial weights derived from a separable cubic convolution kernel. The kernel assigns higher weight to pixels near the sampling location and progressively lower weight to more distant neighbours, which produces smooth transitions without synthesising new high-frequency components.

This behaviour is clinically relevant because all structures in the enlarged image originate from an affine combination of observed intensities. Lesion geometry, vessel curvature and reflection patterns remain unchanged, even though edges may appear softer relative to true high-resolution photographs. The interpolation step produces a magnified fundus image that contains no information not already present in the original acquisition.

From a methodological standpoint, bicubic interpolation establishes a baseline for attribution of improvement. If downstream grading models show higher performance on bicubic-scaled images than on degraded images, then resolution uniformity alone is responsible. If learnable SR models exceed bicubic performance while preserving morphology, the gain can be interpreted as recovery of attenuated retinal detail rather than invention of new texture. This aligns with observations in recent fundus enhancement research that caution against equating perceptual sharpness with diagnostic truth [24].

Bicubic interpolation therefore serves two roles within this study: it provides a clinically safe enlargement path that cannot generate false lesions, and it defines a benchmark that separates genuine high-frequency feature recovery from changes in appearance that are not grounded in retinal anatomy.

3.3.1.2 SRCNN

SRCNN represents one of the earliest learnable super-resolution architectures and forms the conceptual basis for later CNN-based SR models. Unlike bicubic interpolation, which magnifies an image through a deterministic kernel, SRCNN learns a direct mapping from blurred image patches to their sharper counterparts using a three-layer convolutional pipeline [30]. The model operates entirely in the spatial domain and contains no adversarial or perceptual loss components, which gives it predictable behaviour that aligns with clinical safety requirements. Mathematically, given an input image I_{bicubic} that has already been upsampled to the target size using bicubic interpolation, SRCNN computes

$$I' = f_3(f_2(f_1(I_{\text{bicubic}})))$$

where f_1 , f_2 and f_3 correspond to sequential convolutional transformations with non-linear activations. The first layer performs high-dimensional patch extraction using a large 9×9 kernel, capturing luminance and chromatic context across the optic disc, macular region and major vessel paths. The second layer performs non-linear feature mapping using a 5×5 kernel to refine boundaries and transitions, including lesion edges and vascular bifurcations. The final 5×5 reconstruction layer converts the feature maps back into RGB space and clamps pixel values to a valid physiological range.

Because SRCNN reconstructs detail from features learned on top of bicubic upsampling, every restored structure is bounded by intensities that originally existed in the input image. Vessel calibre, foveal curvature and microaneurysm boundaries may appear sharper, but their spatial configuration is not altered. This stands in contrast to GAN-based perceptual SR models that encourage texture synthesis and can introduce features resembling haemorrhages or neovascular loops. The absence of adversarial objectives in SRCNN prevents this failure mode and ensures that improved visibility does not come at the expense of diagnostic trust.

For diabetic retinopathy imaging, this behaviour has important implications. Systematic reviews of fundus enhancement note that SRCNN enhances the clarity of vascular crossings and lesion silhouettes without altering retinal topology [24]. In practice, SRCNN tends to increase the salience of low-contrast abnormalities (for example small haemorrhages) that may be lost under blur or downsampling while preserving global structure such as vessel branching patterns. As a result, SRCNN provides a clinically acceptable form of super-resolution that improves visibility without fabricating new pathological cues.

Within the experimental design of this study, SRCNN acts as the highest-capacity model that remains clinically conservative. If downstream grading models outperform the bicubic baseline but do not exhibit behaviours associated with over-enhancement, the gain can be attributed to the recovery of true fine-scale fundus detail rather than synthetic texture. Thus, SRCNN serves both as a visibility enhancer and as a checkpoint between interpolation-based SR and more generative SR techniques that were intentionally excluded for safety reasons.

3.3.1.3 FSRCNN

FSRCNN was introduced as a computationally efficient reformulation of SRCNN, designed to reduce inference time while preserving the structural consistency of the reconstructed image [31]. Whereas SRCNN applies convolutional refinement after bicubic upsampling, FSRCNN inverts this pipeline and performs nearly all computation in the low-resolution domain before a single learnable upsampling layer. This architectural decision has significant implications for medical imaging, as improvements in visibility are obtained without a proportional increase in computational load or the risk of texture hallucination. Formally, given a low-resolution fundus image I_{LR} , FSRCNN computes

$$I' = g_{\text{deconv}} \left(g_{\text{expand}} \left(g_{\text{map}} \left(g_{\text{shrink}} \left(g_{\text{feat}}(I_{LR}) \right) \right) \right) \right),$$

where g_{feat} performs feature extraction with a large receptive field, g_{shrink} compresses channel dimensionality, g_{map} applies multiple non-linear transformations in the bottleneck space, g_{expand} restores channel dimensionality, and g_{deconv} performs learnable upsampling. The key distinction is that most computations are executed at low spatial resolution, reducing memory footprint and runtime.

The shrinking and mapping stages are particularly relevant to retinal imaging because they learn compact encodings of curvature, luminance gradients and local lesion contrast before magnification. This produces sharper reconstructions of microvascular junctions, haemorrhage borders and reflective highlights on the optic disc without modifying their spatial configuration. Since the final upsampling is learnable, FSRCNN can better reverse blur than bicubic interpolation while remaining bounded to observed input intensities. As a result, it does not invent new structures, in contrast to GAN-based SR models that may synthesise vessel-like edges or pseudo-exudates.

From a clinical deployment perspective, FSRCNN has two practical advantages. First, its low-resolution feature processing makes it substantially faster than SRCNN for comparable output sizes, allowing real-time or near-real-time use on embedded hardware and rural screening stations where GPU resources are limited. Second, the bottleneck representation reduces the

risk of amplification of noise, helping to ensure that increased sharpness comes from authentic structural recovery rather than high-frequency artefacts.

Previous work in fundus enhancement has noted that lightweight CNN-based SR is preferable for diagnostic applications because it preserves vascular topology and lesion geometry [10]. In this study, FSRCNN occupies a methodological midpoint between bicubic interpolation and higher-capacity CNN SR: it offers greater restoration capability than SRCNN while maintaining behaviour that is clinically conservative and predictable. If downstream diabetic retinopathy grading improves under FSRCNN relative to bicubic, and without evidence of misclassification due to over-sharpened artefacts, then the improvement can be attributed to enhanced visibility rather than hallucination of pathology.

3.3.1.4 ESPCN

The Efficient Sub-Pixel Convolutional Network (ESPCN) performs super-resolution using a fundamentally different strategy from SRCNN and FSRCNN. Instead of refining detail after interpolation, ESPCN learns feature representations entirely in the low-resolution domain and defers spatial magnification to the final stage [32]. This architecture is clinically relevant because feature extraction occurs before any enlargement of the sensor grid, reducing the likelihood of synthesising high-frequency texture that could resemble pathology. Let I_{LR} denote the low-resolution input fundus image. Structural features are first extracted by a sequence of convolutional layers:

$$F_1 = \sigma(W_1 * I_{LR} + b_1), \quad F_2 = \sigma(W_2 * F_1 + b_2),$$

where $*$ denotes convolution and $\sigma(\cdot)$ is a non-linear activation. These features encode luminance gradients, vessel geometry and local lesion shape without altering spatial layout. Upscaling is performed only at the final stage. The last convolution predicts r^2C feature channels for an upscale factor r , after which the pixel-shuffle operator $\mathcal{P}$ rearranges them deterministically into spatial coordinates:

$$I_{SR} = \mathcal{P}(W_3 * F_2 + b_3).$$

The pixel-shuffle transformation converts channel depth into spatial resolution without producing new edge patterns. As a result, each output pixel corresponds to a structured subset of learned low-resolution features rather than a hallucinated structure. In retinal imaging, this behaviour improves the visibility of subtle vascular bifurcations, microaneurysm boundaries and haemorrhage margins while maintaining retinal topology. This avoids the pathology-like artefacts reported when adversarial training is used in image enhancement [33]. Because most computation occurs on the low-resolution grid, ESPCN also provides the fastest inference among

the evaluated SR models, which supports practical deployment in screening pipelines where computational resources may be constrained.

Together, these properties make ESPCN an appropriate form of clinically conservative super-resolution: it amplifies anatomical detail already present in the degraded image and commonly does not introduce new structures that could disturb diagnostic interpretation on larger scale reconstruction [10, 24].

3.3.1.5 Summary of selected SR models and clinical rationale

The four SR models form a scale of clinically conservative enhancement, beginning with bicubic interpolation as a non-learned baseline and progressing through SRCNN and FSRCNN, which learn mappings from blurred to sharper patches without adversarial objectives, and ESPCN, which further improves efficiency while maintaining structural fidelity [29–32]. Their shared design priority is to enhance visibility without altering retinal anatomy, consistent with clinical guidance warning that artificial high-frequency texture can mimic pathology and undermine diagnostic reliability [10, 24, 33].

Model	Where computation occurs	Clinical behaviour	Computational cost
Bicubic interpolation	No learning; polynomial interpolation on a 4×4 pixel neighbourhood	No sharpening or texture synthesis; lesion and vessel geometry fully preserved	Lowest (no model required)
SRCNN	All operations after bicubic upsampling (high-resolution space)	Sharpens lesions and vessels without changing morphology; no adversarial loss	Moderate (three-layer CNN)
FSRCNN	Most operations in low-resolution space; deconvolution for final upsampling	Improves vessel crossings and lesion contours without hallucination; topology preserved	Lower than SRCNN due to bottleneck mapping
ESPCN	Fully in low-resolution space; sub-pixel convolution for final reconstruction	Enhances fine detail while preventing texture synthesis; strong safety for diagnosis	Lowest of learnable models; real-time capable

Table 3.2: Super-Resolution Model Summary of Computation and Clinical behaviour

Taken together, these architectures prioritise the preservation of retinal structure over perceptual sharpness. Bicubic marks the upper limit of completely risk-free enhancement. SRCNN and FSRCNN introduce controlled learnability while maintaining predictable clinical behaviour. ESPCN maximises efficiency and structural faithfulness for deployment in settings where computational resources are limited. By restricting the SR design space to these clinically conservative models, the study aims to make any observed improvement in diabetic retinopathy severity detection primarily attributable to the recovery of genuine anatomical detail rather than the synthesis of visually appealing texture.

3.3.2 SR inference and output normalisation

Super-resolution is only clinically meaningful when the output remains statistically compatible with a real fundus photograph. For this reason, all super-resolved images were forced to match the original spatial resolution and photometric distribution of the dataset before being supplied to downstream graders. This ensures that any improvement in classification accuracy arises from the recovery of fine-scale retinal detail rather than from differences in colour balance or exposure. During inference, the SR output $\hat{I}$ was normalised using three sequential constraints. These constraints were applied identically to bicubic, SRCNN, FSRCNN and ESPCN:

- **Histogram mean alignment:** the global intensity mean of $\hat{I}$ was shifted to match the original image mean μ_I on a per-channel basis.
- **Variance stabilisation:** pixel variance of $\hat{I}$ was corrected to fall within the physiological variance range observed in the dataset.
- **Value clipping:** the output was confined to $[0, 1]$ to prevent unrealistic sensor responses.

Normalisation Component		Motivation in DR Screening
Histogram alignment	mean	Prevents SR from appearing to enhance pathology by increasing brightness or saturation. Ensures original illumination context is preserved.
Variance stabilisation		Stops artificial contrast amplification that could cause benign tissue to resemble haemorrhages or exudates.
Value clipping $[0, 1]$		Prevents pixel values outside physiological fundus intensity range, particularly in specular highlights and optic disc regions.

Table 3.3: Photometric normalisation components applied uniformly to all SR outputs.

Following these constraints, super-resolved images retain the same global appearance as the originals while recovering local anatomical detail, such as lesion edges and vascular curvature. Consequently, improvements in diabetic retinopathy grading can be attributed to restored visibility rather than to altered illumination or contrast.

3.3.3 Runtime and compute efficiency considerations

The SR models were selected with deployment in screening environments rather than laboratory hardware. All learnable models in this study (SRCNN, FSRCNN and ESPCN) operate within the memory and throughput limits of mid-range GPUs and edge computing devices commonly available in ophthalmic clinics, mobile vans and rural tele-ophthalmology programs. None of the models require discriminator networks, transformer backbones or perceptual loss networks, which avoids significant compute and inference overhead.

FSRCNN and ESPCN offer additional efficiency by performing most computation in the low-resolution space. FSRCNN avoids convolution in the expanded feature map by applying learnable upsampling only at the final stage. ESPCN performs resolution increase via a sub-pixel pixel-shuffle operation rather than through deconvolution. This makes both models suitable for continuous high-volume inference.

SR Model	Main Compute Domain	Upsampling Method	Clinical Deployment Suitability
Bicubic	Full-resolution	Deterministic kernel	Very high (no computation overhead)
SRCNN	Full-resolution	Convolution-based reconstruction	High (mid-range GPU)
FSRCNN	Low-resolution	Learnable deconvolution at final stage	Very high (low compute load)
ESPCN	Low-resolution	Pixel-shuffle (sub-pixel rearrangement)	Very high (edge-device compatible)

Table 3.4: Compute strategies of SR models and their suitability for clinical screening deployment.

Overall, the computational characteristics of the chosen SR models align with the constraints of real-world ophthalmic screening, where reliability and throughput are prioritised over maximum sharpness. This ensures that SR can be applied routinely in environments where GPU resources, bandwidth and time are limited.

3.4 Supervised Baseline Classifiers

3.4.1 Classical ML classifiers (Logistic Regression, SVM, Random Forest, XGBoost)

Classical machine learning classifiers were included to establish a low-capacity baseline that does not learn directly from pixel distributions. All models operated on frozen deep feature embeddings extracted from the same backbone, ensuring that their behaviour reflects changes in lesion separability rather than improvements in image appearance.

3.4.1.1 Logistic Regression

Multinomial Logistic Regression provides the simplest decision boundary among the baselines. For class logits $z_k = w_k^\top x + b_k$, class probabilities are

$$P(y = k \mid x) = \frac{\exp(z_k)}{\sum_{j=1}^K \exp(z_j)}.$$

Because Logistic Regression relies on linear separability, it typically shows the earliest performance decline when visibility is reduced. If performance improves under SR without retraining, the gain can be attributed to an increase in global separability of severity patterns in the embedding space.

3.4.1.2 Support Vector Machine (RBF)

The SVM baseline tests whether non-linear margins in the embedding space remain stable when visibility degrades. Its decision function

$$f(x) = \text{sign} \left(\sum_{i=1}^N \alpha_i y_i \exp(-\gamma \|x - x_i\|^2) + b \right)$$

allows fine-grained discrimination between neighbouring DR grades, but margins collapse rapidly when small lesions lose contrast. SVM therefore acts as a sensitive indicator of visibility loss.

3.4.1.3 Random Forest

Random Forest provides a noise-tolerant baseline. It aggregates the mode of T decorrelated decision trees,

$$\hat{y} = \text{mode}(h_1(x), h_2(x), \dots, h_T(x)),$$

which stabilises predictions when embeddings become unreliable. RF typically declines more slowly than SVM or Logistic Regression under degradation, but does not strongly benefit from SR when improvements are subtle.

3.4.1.4 XGBoost

XGBoost represents the highest-capacity classical model, learning residual refinements across boosted trees,

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i),$$

optimised under a regularised objective

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \left(\lambda \|w_k\|^2 + \gamma \text{num_leaves}(f_k) \right).$$

Because XGBoost can model small gradations between DR severities, it provides the strongest indication of whether SR recovers clinically meaningful structure rather than only improving contrast.

Model	Strength in DR embedding space	Impact of visibility loss
Logistic Regression	Tests global linear separability of DR features	First to collapse; improves only if SR increases global separability
SVM (RBF)	Captures fine lesion boundaries through local kernel geometry	Steep decline under blur; recovery only if SR restores subtle structure
Random Forest	Stable when embeddings are noisy or partially ambiguous	Slow decline; limited recovery from SR unless structural cues are large
XGBoost	Fits incremental refinements of neighbouring severity grades	Most resilient; strong recovery only if SR restores clinically relevant cues

Table 3.5: Classical machine learning baselines and their diagnostic interpretation under degradation and super-resolution conditions.

Because none of these models were trained on SR images, any improvement during evaluation reflects an increase in the discriminability of true structural information rather than an aesthetic change in appearance. Classical classifiers therefore provide a strict control for interpreting the behaviour of CNN and ZSL methods later in the pipeline.

3.4.2 Deep CNN and Vision Transformer baselines (ResNet, EfficientNet, ViT)

Supervised deep learning models were incorporated to establish a high-capacity reference point for diabetic retinopathy classification. Unlike classical ML baselines, which operate on fixed feature embeddings, these networks learn directly from pixel intensities and therefore serve as an upper bound on the benefit that super-resolution (SR) can provide to automated grading.

3.4.2.1 ResNet

ResNet models were selected as canonical convolutional baselines due to their stable optimisation behaviour and hierarchical representation of retinal structures. A ResNet block computes

$$y = F(x, \mathcal{W}) + x,$$

where the residual function F refines features from the previous layer rather than replacing them. This formulation expands the effective depth of the network without increasing gradient instability.

In retinal analysis, early ResNet layers typically learn colour and illumination trends, intermediate layers model vessel trajectories and optic disc boundaries, and late layers encode lesion morphology such as microaneurysms, haemorrhages and exudates. Two depths were employed: ResNet18 as a moderate-capacity model that prioritises low-level texture, and ResNet50 as a high-capacity architecture capable of modelling interactions between multiple lesion types. Under blur and low-contrast degradation, mid-level lesion cues vanish first, causing ResNet performance to decline if SR fails to restore physiologically plausible microstructures.

3.4.2.2 EfficientNet-B0

EfficientNet introduces compound scaling of depth d , width w and input resolution r ,

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi,$$

with constants chosen to maximise accuracy under fixed computational cost. The architecture combines mobile inverted bottleneck convolution (MBConv) with squeeze-and-excitation (SE) attention. MBConv preserves fine spatial detail using depthwise convolution, while SE modules reweight channels according to global contextual statistics.

These properties are particularly well-matched to DR because lesion intensity, size and spatial extent vary substantially across severity grades. EfficientNet-B0 therefore provides a strong reference for clinically valid performance under realistic computing constraints, including com-

munity screening and tele-ophthalmology deployments. Performance is expected to degrade more slowly than ResNet under blur due to the strong texture-sensitive behaviour of MBConv, and to recover moderately if SR reinstates the contrast profile of small lesions.

3.4.2.3 Vision Transformer (ViT-B16)

ViT represents a substantially different modelling paradigm. Each image is decomposed into non-overlapping $P \times P$ patches and embedded into a sequence of tokens that are processed by multi-head self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V.$$

This formulation establishes long-range contextual dependencies between retinal regions, enabling the model to reason jointly over the macula, optic disc and peripheral vascular fields. The inclusion of positional encodings allows ViT to reconstruct the spatial layout of retinal anatomy despite the lack of convolutions.

The global nature of attention makes ViT highly sensitive to the disappearance of small patches containing lesions. Under degradation, tokens corresponding to microaneurysms and haemorrhages may become visually indistinguishable from healthy tissue, causing attention weights to collapse. ViT therefore provides the strictest test of whether SR restores clinically relevant information rather than merely increasing sharpness. If ViT performance improves after SR, it indicates not only local lesion recovery but also restoration of disease-linked anatomical context.

Model	Clinical and architectural strengths	Expected response to degradation and SR
ResNet18 / ResNet50	Multi-scale lesion and vessel representation through progressive residual refinement	Early decline when small lesions blur; recovery occurs only if SR reconstructs physiologically plausible microstructures
EfficientNet-B0	MBConv and SE attention recover high-value textural cues with minimal computational load	More resilient under blur; modest recovery from SR when colour and lesion contrast are reinstated
ViT-B16	Patch-wise global reasoning linking macula, optic disc and vascular tree	Sharp decline under degradation; strong recovery only if SR restores lesion identity at the token level and across spatial context

Table 3.6: Technical rationale for deep supervised baselines and their expected diagnostic behaviour under degradation and SR.

These models collectively define the supervised performance envelope of the study. If SR benefits only ZSL, visibility is not the limiting factor for supervised deep learning. If both deep learning and ZSL recover under SR, lesion visibility is the primary bottleneck across model families.

3.4.3 Training configuration and hyperparameters

All supervised models were trained strictly on original images to preserve internal validity and prevent information leakage from super-resolved inputs. This ensures that any change in performance under degradation or SR arises from altered image visibility rather than from altered supervision or retraining effects. To make comparisons fair across architectures, each model received the same input resolution, the same augmentation pipeline and the same optimisation schedule, removing training configuration as a source of variability.

Parameter	Configuration (shared across all deep-learning baselines)
Training data	Original dataset only (APTOS or IDRiD depending on experiment)
Evaluation splits	Original, degraded and all SR variants on both datasets
Input resolution	224×224 pixels
Normalisation	ImageNet mean / standard deviation
Data augmentation	Random horizontal flip and $\pm 10^\circ$ rotation
Optimiser	Adam, learning rate 1×10^{-4}
Loss function	Cross-entropy loss
Batch size	32
Max epochs	20
Early stopping	Triggered after 5 epochs without val accuracy improvement
Checkpoint selection	Best validation-accuracy checkpoint restored for evaluation
Hardware	GPU when available; CPU as fallback

Table 3.7: Uniform training configuration used for all CNN and ViT models.

This tightly controlled protocol prevents models from adapting to degradation or SR features during training, meaning that any subsequent accuracy recovery after SR reflects genuine restoration of lesion visibility. The configuration also guarantees comparability across architectures: improvements cannot arise from hyperparameters, learning rate schedules or augmentation choices, but only from differences in representational capacity and sensitivity to fine-grained retinal structures.

3.4.4 Role of supervised baselines

Supervised CNN and transformer models serve as high-capacity reference points for determining whether SR benefits arise from restored lesion visibility rather than from shifts in contrast, texture, or dataset statistics. Because these models are trained only on original images and already achieve strong performance without exposure to degraded or SR inputs, their behaviour across conditions operates as a stability anchor. If SR meaningfully restores pathological structure, ZSL performance should respond strongly, while CNN/ViT performance should remain comparatively stable due to their learned invariances.

Reference Signal		Expected behaviour if SR restores visibility	Expected behaviour if SR alters image statistics
ZSL (CLIP-based)		Large gains, particularly for early DR and degraded cases; sensitivity to subtle lesion enhancement	Erratic shifts unrelated to clinical patterns; improvements even when lesions are not present
CNN (ResNet / Efficient-Net)		Little or no change across SR conditions due to invariance to blur/contrast	Noticeable performance fluctuations; shift in confidence profiles or misclassification trends
Vision Transformer (ViT)		Minor or gradual changes; ViTs may capture slightly more fine structure but remain stable overall	Large shifts, especially in high-grade predictions, indicating distribution change rather than restored detail
Classical ML (SVM / RF / XGBoost)		Strong recovery only if embeddings become more separable after SR	Recovery even when lesions are not realistically improved; inconsistent with CNN/ViT behaviour

Table 3.8: Expected diagnostic patterns across model families under two competing hypotheses: SR restores visibility vs SR modifies appearance distribution.

The contrast between ZSL sensitivity and supervised model stability is what enables causal interpretation. When SR improves ZSL on early DR and degraded imagery while CNN/ViT performance remains unchanged, the limiting factor is visibility rather than dataset size or model capacity. In this case, ZSL benefits emerge only when SR recovers lesion structure, whereas supervised baselines remain near-constant because they already encode invariances that buffer against mild blur and luminance changes. Conversely, if supervised networks show large accuracy swings across SR conditions, the effect is likely distributional rather than biomedical, signalling that SR may be modifying appearance statistics rather than restoring lesions. In this study, supervised networks therefore function not simply as additional benchmarks but as an internal validity mechanism that grounds interpretation of ZSL gains.

3.5 Zero-Shot and Prompt-Based Classification

3.5.1 Vision-language model configuration (CLIP and RETCLIP)

Zero shot diabetic retinopathy classification was implemented using vision-language models that embed fundus images and natural language class prompts into the same metric space. When a model is operated in frozen mode, the image encoder and text encoder cannot adjust

to dataset-specific characteristics. Performance differences therefore isolate the effect of retinal visibility rather than optimisation. This makes CLIP and RETCLIP suitable tools for analysing whether super resolution restores clinically relevant structure.

3.5.1.1 CLIP

CLIP learns joint visual and linguistic representations from large scale image-caption pairs by aligning the latent distribution of both modalities using contrastive learning [14]. A visual encoder converts an input image to a feature vector, and a text encoder converts a prompt to another feature vector of equal dimensionality. During training, cosine similarity between matching pairs is maximised while similarity to all mismatched captions is minimised. This procedure results in an embedding space in which semantic proximity reflects conceptual similarity. Because CLIP has never been trained on retinal datasets, its predictions for diabetic retinopathy grades depend exclusively on the recognisability of pathological patterns rather than dataset-specific parameter updates.

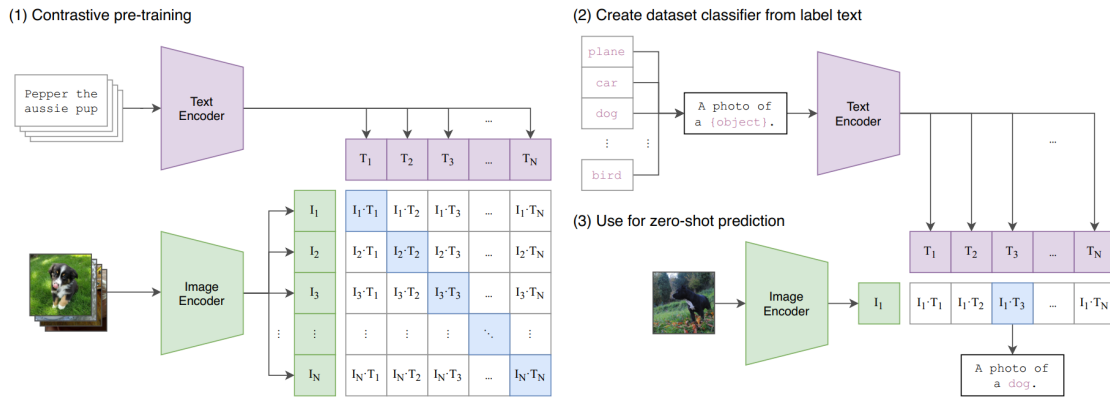


Figure 3.6: CLIP training and zero shot prediction [14].

After training, classification becomes a nearest neighbour problem. The model receives a fundus image and a set of class prompts. Each prompt is embedded into the shared space and the predicted class corresponds to the prompt whose text vector has the maximum cosine similarity to the image vector. Importantly, this mechanism does not depend on gradient updates or labelled DR samples. CLIP therefore acts as a direct probe for assessing the intrinsic visibility of biomarkers such as microaneurysms, haemorrhages and neovascularisation under degradation and super resolution.

3.5.1.2 RETCLIP

RETCLIP modifies the CLIP pipeline to address domain shift between natural photography and retinal imaging [34]. Instead of learning from generic image-caption pairs, RETCLIP uses fundus photographs paired with clinical diagnostic reports. The text encoder remains unchanged relative to CLIP, but the visual encoder is retrained so that retinal structures become the dominant contributors to the metric space. This adjustment reduces sensitivity to lighting variation and camera hardware differences while increasing sensitivity to macular geometry, vascular patterns and lesion morphology.

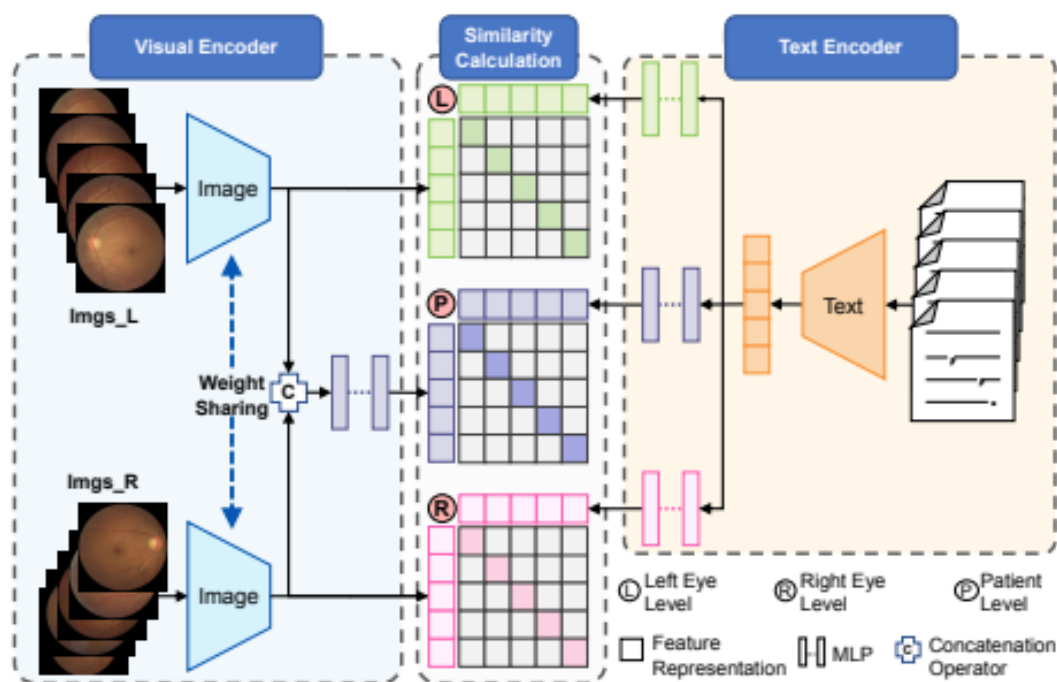


Fig. 1. Overview of the RET-CLIP foundation model.

Figure 3.7: RETCLIP visual-text alignment for retinal images [35].

Because RETCLIP is adapted to the retinal modality, it is expected to rely more heavily on fine grained biomarkers. In practice, RETCLIP often fails on low visibility inputs where pathological cues are partially obscured or blurred. If super resolution restores structural detail, RETCLIP accuracy increases sharply, whereas CLIP may improve primarily from global visibility changes such as contrast recovery. The contrasting behaviours of the two models therefore create a diagnostic signal. When CLIP improves but RETCLIP does not, super resolution has restored generic visibility without restoring pathological detail. When RETCLIP improves, lesion morphology has been reconstructed at a clinically meaningful level. Tracking both models enables the study to separate cosmetic image improvements from genuine disease relevant restoration.

3.5.1.3 Comparison between CLIP and RETCLIP

Although both models operate within the same image-text alignment framework, their behaviour differs substantially when applied to diabetic retinopathy. CLIP reflects general semantic visibility because its visual encoder has never been exposed to retinal datasets. RETCLIP, in contrast, reflects the recoverability of domain specific biomarkers because its visual encoder has been adapted to clinical retinal photography. This contrast allows the two models to serve complementary analytical roles in the evaluation of super resolution.

Property	CLIP	RETCLIP
Training source	Generic natural-image and caption pairs	Fundus images paired with clinical diagnostic reports
Visual encoder domain	General photography	Retinal imaging
Text encoder	Large scale language encoder (unchanged in both models)	Same text encoder as CLIP
Zero-shot mechanism	Cosine similarity between image and prompt embeddings	Same mechanism, but similarity emphasises retinal structures
Sensitivity to visibility loss	Declines gradually due to global blur and contrast loss	Declines sharply when fine grained biomarkers disappear
Expected behaviour under SR	Moderate improvement when general visibility is restored	Large improvement when lesion morphology is reconstructed
Interpretation of gains	Recovery of structural legibility	Recovery of clinically meaningful features
Analytical role in this study	Measures generic visibility restoration	Measures diagnostic signal restoration

Table 3.9: Conceptual comparison of CLIP and RETCLIP in the context of zero shot diabetic retinopathy classification.

3.5.2 Rationale for zero-shot classification in diabetic retinopathy

Zero-shot learning (ZSL) offers an alternative to supervised diabetic retinopathy (DR) classification by enabling disease staging from natural-language prompts rather than annotated retinal datasets. Unlike deep CNN and transformer models, whose performance scales primarily with training set size and ground-truth label quality [1, 2, 22], ZSL derives its predictions from the similarity between an image embedding and text embeddings that describe disease categories in a shared semantic space [14]. This eliminates the need for expert-labelled retinal datasets, which remain costly, time-consuming and inconsistent across imaging devices, populations and clinical grading conventions [4, 16].

The suitability of ZSL for DR research stems from three factors. First, because ZSL models are not optimised on DR images, their predictions reflect the intrinsic recognisability of retinal pathology rather than successful gradient-based learning. When a ZSL model distinguishes between mild and moderate DR purely from text prompts and image features, it demonstrates that relevant biomarkers are visibly present rather than memorised during supervised training. Second, ZSL prevents annotation density from confounding the evaluation of image super-resolution. Supervised performance improves as the number and quality of annotations increase, whereas ZSL performance depends almost exclusively on the visibility and semantic alignment of retinal lesions [21, 36]. Third, the ZSL paradigm mirrors deployment settings in which annotated data is scarce but ophthalmic knowledge is widely available in clinical language, which is especially relevant for low-resource or early-deployment screening scenarios [6, 7].

Within this study, ZSL therefore functions as a visibility-sensitive diagnostic probe. If super-resolution improves ZSL performance on degraded fundus photographs, the recovered information must correspond to clinically meaningful retinal structures rather than generic contrast sharpening. If ZSL performance does not recover under SR, the reconstructed pixels are unlikely to encode genuine biomarkers. This makes ZSL an effective mechanism for determining whether super-resolution enhances disease information rather than only visual sharpness, without the confounding interaction between dataset scale, model capacity and annotation quality that affects supervised approaches.

3.5.3 Prompt engineering and paraphrasing strategy

Zero-shot performance is inherently sensitive to linguistic phrasing because image-text alignment relies on cosine similarity between an image embedding and the semantic meaning encoded within a language prompt rather than a fixed class index [14]. Even clinically equivalent labels, such as “moderate diabetic retinopathy” and “Stage 3 non-proliferative DR”, may activate different representations depending on the text encoder’s training corpus [21, 36]. To mitigate linguistic bias and prevent prompt phrasing from artificially affecting classification, each disease category was represented by a set of paraphrased prompts designed to preserve semantic identity while introducing lexical and syntactic diversity.

3.5.3.1 Prompt construction principles

Zero-shot fundus classification is highly sensitive to linguistic phrasing. Small changes in wording can alter text-embedding geometry and therefore shift cosine similarity scores that drive prediction. To minimise prompt-induced variance and to ensure that performance reflects retinal visibility rather than linguistic artefacts, prompt sets were constructed using four complementary design principles grounded in the DR literature and prior medical VLM research [17, 18].

1. **Grounded clinical terminology:** terms that explicitly reference pathological structures such as microaneurysms, cotton wool spots and neovascular tissue. These cues benefit models that rely on lesion-level semantic priors.
2. **Grading-aligned phrasing:** sentences aligned to severity scales defined by APTOS and IDRiD, for example “mild non-proliferative diabetic retinopathy” or “very severe NPDR” [5, 27]. This format exploits the fact that some VLMs encode ordinal disease progressions.
3. **Paraphrasing to reduce encoder bias:** multiple syntactic variants for the same class (short-form, descriptive form, paragraph-style form) to prevent over-reliance on a single lexical pattern. This reduces the risk that a model performs well only when the prompt matches its preferred phrasing template.
4. **Balanced sentiment across disease classes:** neutral wording that avoids emotionally charged or subjective adjectives. This prevents text embeddings from acquiring separability based on sentiment polarity rather than pathology.

These four principles serve different purposes rather than acting redundantly. Clinical-terminology prompts test whether the model recognises anatomical markers, grading-aligned prompts assess whether ordinal DR progression is encoded textually, paraphrased prompts probe linguistic

challenges, and sentiment-balanced prompts ensure that decision boundaries arise from retinal structure rather than stylistic cues. Together, they form a prompt set that is semantically precise while minimising the risk of spurious linguistic advantages.

3.5.3.2 Dataset-specific prompt considerations

Because IDRiD and APTOS define severity categories differently, prompts were adapted to match each dataset’s clinical taxonomy without introducing semantic drift. IDRiD follows the five-tier DR staging system (No DR, Mild, Moderate, Severe, and Proliferative) [5], whereas APTOS uses clinical descriptions with intermediate separation between moderate and severe classes [27]. A single unified prompt template would have conflated disease groups across datasets and distorted embeddings, therefore dataset-specific prompt sets were developed.

Dataset	Severity taxonomy reflected in prompts	Example paraphrased prompt set
IDRiD	Five-class staged DR scale	<i>"Moderate diabetic retinopathy"; "Stage 3 non-proliferative diabetic retinopathy"; "Presence of multiple microaneurysms and haemorrhages without neovascularisation"</i>
APTOS	Broader class definitions with expanded clinical descriptors	<i>"Severe diabetic retinopathy with extensive haemorrhages"; "Late-stage non-proliferative diabetic retinopathy"; "Very severe DR with extensive vascular abnormalities"</i>

Table 3.10: Differences in prompt phrasing strategy between IDRiD and APTOS due to varying severity taxonomies.

3.5.3.3 Motivation for paraphrasing in the DR context

Prompt paraphrasing maximises the stability of zero-shot classification by ensuring that prediction shifts reflect changes in lesion visibility rather than isolated text-embedding interactions. In the context of this study, any improvement in performance across paraphrased prompts provides stronger evidence that super-resolution enhanced the visibility of DR pathology rather than matched a specific wording. Conversely, if predictions diverge heavily across prompts, the underlying retinal features are unlikely to be sufficiently pronounced to override linguistic variation. This makes prompt diversity a required control.

3.5.4 Embedding similarity and decision strategy

Zero-shot diabetic retinopathy classification relies on comparing the embedding produced by the visual encoder to the embeddings produced by each class prompt. For an input image I , the visual encoder produces an embedding $v \in \mathbb{R}^d$ and the text encoder produces a set of prompt embeddings $\{t_1, t_2, \dots, t_C\}$ for the C candidate classes. Both vectors are L2 normalised so that their dot product corresponds to cosine similarity:

$$\text{sim}(v, t_c) = \frac{v \cdot t_c}{\|v\| \|t_c\|}.$$

The class prediction is obtained by selecting the prompt with the highest similarity:

$$\hat{y} = \arg \max_{c \in \{1 \dots C\}} \text{sim}(v, t_c).$$

When multiple paraphrased prompts represent the same class, their similarity scores are averaged within class before the final decision. For a class with K paraphrases, the similarity score is

$$S_c = \frac{1}{K} \sum_{k=1}^K \text{sim}(v, t_{c,k}),$$

and the predicted class becomes

$$\hat{y} = \arg \max_{c \in \{1 \dots C\}} S_c.$$

This ensembling strategy mitigates prompt-specific sensitivity and ensures that decisions arise from consistent semantic alignment rather than a single preferred phrasing pattern. Importantly, no temperature scaling, calibration or post-hoc thresholding was applied, which preserves internal validity by ensuring that changes in accuracy reflect differences in retinal visibility rather than hyperparameter optimisation.

Component	Role in decision process
Visual embedding v	Encodes retinal structure and reflects the effect of super-resolution and degradation on lesion visibility.
Text embeddings $\{t_c\}$	Encode the clinical semantics and disease-stage descriptors expressed in the prompt set.
Cosine similarity	Computes the geometric alignment between retinal features and diagnostic language in the shared embedding space.
Prompt ensembling	Averages similarity across paraphrased prompts to limit linguistic bias and improve reliability.
Argmax classification	Selects the class with the highest similarity score without temperature scaling or threshold calibration.

Table 3.11: Expanded summary of the similarity scoring and decision process underlying zero-shot classification.

The embedding similarity mechanism therefore serves not only as a numerical scoring rule but as an implicit measure of how “clinically aligned” a retinal image is with the semantic description of each disease stage. Cosine similarity captures the geometric proximity between visual features and diagnostic language in the shared latent space, meaning that a prediction is made only when the retinal appearance and the linguistic description occupy a similar region of the embedding manifold. The decision strategy is deliberately minimal: there is no confidence calibration, no class-specific thresholds and no tuning of temperature parameters. Instead, the classifier selects the disease class whose prompt embedding demonstrates the strongest geometric agreement with the visual embedding. This design preserves internal validity in the context of super-resolution benchmarking, because any change in accuracy reflects an actual change in the recognisability of pathology rather than optimisation of the decision rule. In other words, if super-resolution improves classification, it is because lesions became easier to represent in the shared embedding space, not because the model was adapted to the task.

3.5.5 Binary vs five-class taxonomy justification

Zero-shot diabetic retinopathy classification was evaluated using two taxonomic regimes that reflect complementary clinical objectives. The binary configuration distinguishes healthy from pathological cases and therefore simulates first-line screening where the principal decision is whether a patient requires referral. The five-class configuration corresponds to the canonical progression of diabetic retinopathy and is intended to evaluate whether the model can recognise subtle differences in lesion severity and vascular disruption. These two regimes were included not only for clinical relevance but also because they impose different stresses on the embedding

space used by zero-shot models. Binary classification depends mainly on the detectability of any abnormality, whereas five-class staging depends on the visibility of fine-grained lesion patterns across the disease continuum. Assessing both regimes is therefore essential to determine whether performance gains following super-resolution arise from improved retinal visibility rather than prompt bias or class structure simplification.

As, Binary taxonomy for screening-style deployment was vital, the binary task considers all disease stages from mild retinopathy to proliferative retinopathy are grouped into a single class. The decision boundary therefore depends heavily on the detectability of early pathological cues such as microaneurysms, isolated dot-blot haemorrhages and faint exudates. These are precisely the features that tend to disappear when resolution or image quality decreases. Binary tasks also highlight the sensitivity of zero-shot classification to dataset acquisition conditions, since APTOS and IDRiD differ in colour calibration, illumination and field-of-view. This required dataset-specific prompt phrasing to minimise the effect of domain shift independent of the visual encoder. Representative prompts are summarised in Table 3.12.

Class	Binary (IDRiD) prompt style	Binary (APTOS) prompt style
no_dr	“A healthy retinal fundus image with no signs of diabetic retinopathy.”	“A healthy retinal fundus image with no signs of diabetic retinopathy in this retinal image.”
	“Fundus photograph showing no abnormalities.”	“Fundus photograph showing no abnormalities, intact vessels and optic disc.”
dr	“A retinal image showing signs of diabetic retinopathy.”	“A retinal fundus image showing signs of diabetic retinopathy.”
	“Diabetic retinopathy with visible microaneurysms and abnormalities.”	“Fundus photograph with visible microaneurysms, haemorrhages or exudates.”

Table 3.12: Comparison of binary prompt sets for IDRiD and APTOS. The class definitions are shared but the phrasing is adapted to minimise domain shift between datasets.

The five-class regime evaluates whether the model can distinguish among the recognised stages of diabetic retinopathy rather than simply detecting disease presence. In contrast to the binary task, the five-class setting depends on whether the embedding space retains information about the progressive changes in lesion burden, vascular deformation and neovascular tissue. This makes five-class classification far more sensitive to the visibility of subtle structural cues that are prone to degradation and that super-resolution is intended to recover. While the binary prompts in Table 3.12 concentrate primarily on differentiating healthy from pathological imagery, the

five-class prompt set in Table 3.13 expresses a gradual transition in disease severity, encouraging the model to reproduce the ordinal structure of the clinical taxonomy rather than treating severity as an arbitrary categorical label.

Class	Representative prompt phrasing
no_dr	“Fundus image with no diabetic retinopathy.”
mild	“Mild non-proliferative diabetic retinopathy with few microaneurysms.”
moderate	“Moderate diabetic retinopathy with microaneurysms and dot-blot haemorrhages.”
severe	“Severe non-proliferative diabetic retinopathy with numerous haemorrhages and venous beading.”
proliferative	“Proliferative diabetic retinopathy with neovascularisation.”

Table 3.13: Prompt set used for five-class zero-shot staging, representing the full diabetic retinopathy grading taxonomy.

The contrasting taxonomies also serve an interpretative function when evaluating the impact of super-resolution on zero-shot inference. If improvements occur only in the binary regime, they likely reflect better detectability of overt pathology, aligning with the simpler decision task and the broader phrasing of the binary prompts. However, if performance gains extend to the five-class regime, then the restored resolution is not only revealing the presence of disease but also reinforcing the discriminability of fine-grained visual patterns that correspond to clinically meaningful stages. The paired prompt tables therefore form a deliberate diagnostic tool: they reveal whether super-resolution enhances decision-making by supporting deeper retinal understanding or whether improvements are limited to coarse screening performance.

3.6 End-to-End Experimental Design

3.6.1 Sequential SR and ZSL pipeline

The experimental pipeline was designed to test whether improvements in retinal visibility produced by super-resolution transfer into higher classification accuracy across multiple learning paradigms. Each image existed in two foundational states: (i) the original clinical-resolution fundus image and (ii) a degraded counterpart generated through downsampling, Gaussian blur and Gaussian noise. Three SR models were then applied to both baselines, ESPCN [32], FSRCNN [31] and SRCNN [30] and generating six enhanced variants. Bicubic upsampling [29] served as a non-learned reference, resulting in eight imaging conditions per dataset.

No classifier was ever trained on SR or degraded imagery. All classical ML, deep learning and zero-shot learning models were trained exclusively on the original clinical images and then evaluated on original, degraded and super-resolved variants. This design ensured that any change in performance could be attributed to visibility rather than optimisation on synthetic textures or adaptation to a modified distribution. Figure 3.8 illustrates the full sequential flow from degradation through SR to downstream evaluation.

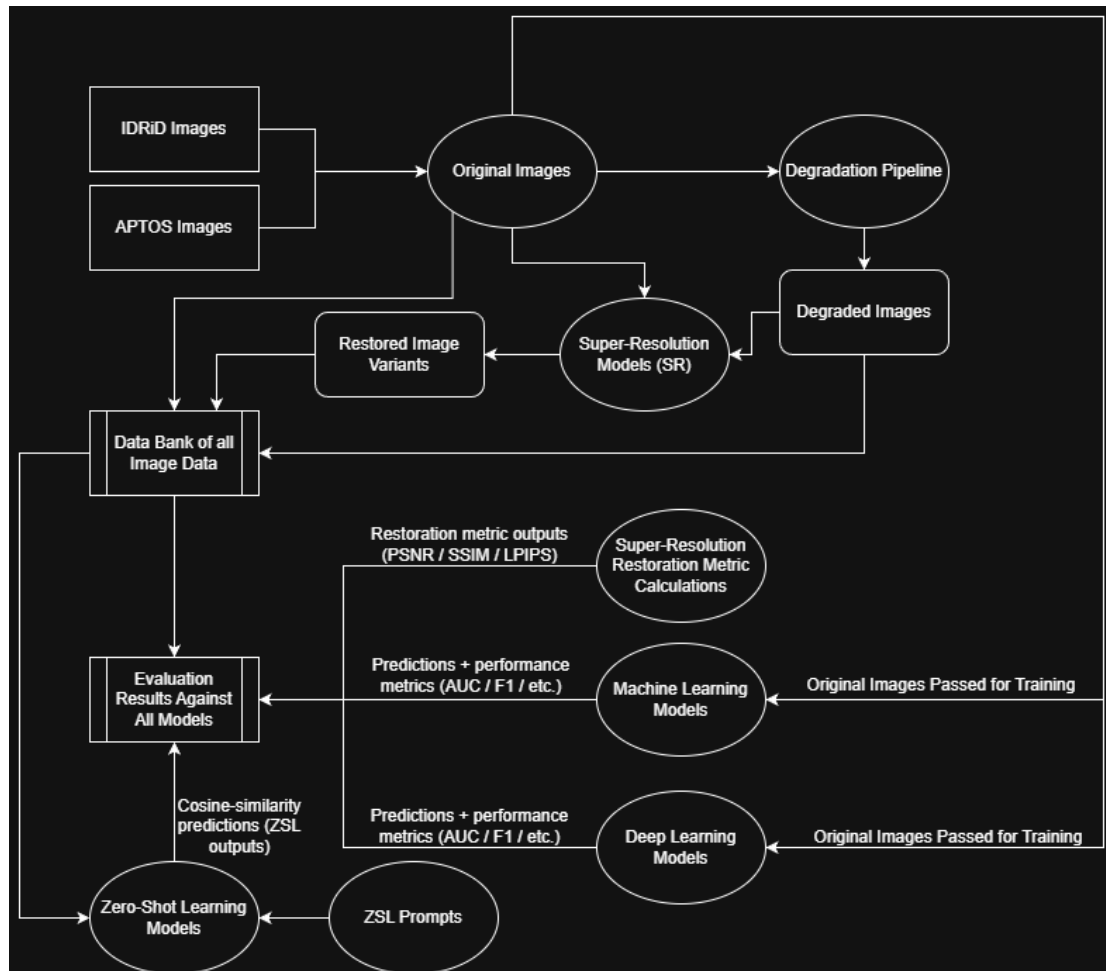


Figure 3.8: Sequential pipeline used throughout the study. All learning-based SR occurs before classification, and no classifier was trained on super-resolved inputs, ensuring causal interpretability of visibility effects.

3.6.2 Independent and dependent variables

Independent variables captured the components deliberately manipulated in the study: dataset (APTOS [27] vs IDRiD [5]), imaging state (original, degraded, bicubic, ESPCN, FSRCNN and SRCNN) and learning paradigm (classical ML, supervised DL and zero-shot learning). Dependent variables represented classification outcomes rather than optimisation convenience, consisting of accuracy, balanced accuracy and F1-weighted for all ML and DL models, and the same metric structure for zero-shot classification.

Variable category	Definition and rationale
Dataset	APTOS [27] and IDRiD [5] selected for complementary imaging styles and annotation protocols.
Imaging conditions	Original, degraded, bicubic, ESPCN [32], FSRCNN [31], SRCNN [30].
Learning paradigms	Classical ML, supervised DL and zero-shot vision-language classification.
Performance metrics	Accuracy, balanced accuracy and F1-weighted reported on all conditions.
Evaluation rule	All models applied to every imaging variant to expose where visibility transfer effects arise.

Table 3.14: Independent and dependent variables used throughout the experimental design.

3.6.3 Controlled factors and confounders

To maintain internal validity, potentially confounding components were fixed throughout the pipeline. Supervised ML and DL models were trained exclusively on original clinical images to avoid advantaging SR-enhanced examples. ZSL models (CLIP [14] and RET-CLIP [34]) were used in fully frozen mode and never fine-tuned on retinal datasets, ensuring that performance reflected the recognisability of pathology rather than adaptation. Prompt sets were fixed per task and were not re-optimised for any dataset or enhancement condition. All images were normalised to the same spatial resolution and pixel-intensity scale prior to classification.

Controlled factor	Justification
Training distribution	All ML/DL models trained exclusively on original images to prevent SR-dependent memorisation.
ZSL model state	CLIP [14] and RET-CLIP [34] kept frozen to ensure that accuracy depends solely on visibility.
Prompt sets	Fixed prompts per task to avoid accuracy inflation through prompt engineering.
Image normalisation	Identical spatial and photometric scaling to prevent brightness- or resolution-based bias.
Evaluation consistency	All imaging variants tested with identical model checkpoints and thresholds.

Table 3.15: Controlled components introduced to prevent confounding of SR-related visibility effects.

3.6.4 Expected vs observed trends

The hypothesis motivating the design was that visibility should matter most when lesion structures are small, sparse or early in their clinical emergence. Early DR often presents with subtle microaneurysms and fine haemorrhagic traces that rapidly disappear under degradation and therefore should be first restored by super-resolution [1, 2]. Moderate and severe DR exhibit widespread vascular disruption that remains visible even under degradation, acting as internal controls where super-resolution would be expected to confer limited benefit.

If performance improvements arise only under binary classification, then visibility restoration primarily supports coarse screening and referral. If improvements extend into the five-class taxonomy, then SR contributes to fine-grained lesion recognition rather than merely enhancing gross abnormality detection. This dual evaluation regime therefore provides a structured way to separate genuine visibility-driven gains from effects caused by prompt design, class simplification or dataset imbalance. Empirical results in the next section test these expectations quantitatively across all models and datasets.

3.7 Evaluation Strategy for Visibility Enhancement

3.7.1 Image-level metrics (PSNR and SSIM)

Visibility enhancement was quantified using two metrics computed for every original-enhanced image pair: peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). These were selected because they jointly capture pixel fidelity and retinal structure preservation without assuming any downstream task model. PSNR measures reconstruction fidelity by comparing the logarithmic ratio between the maximum possible pixel value and the mean squared error between the original clinical-resolution fundus image I and its enhanced counterpart $\hat{I}$:

$$\text{PSNR}(I, \hat{I}) = 10 \log_{10} \left(\frac{255^2}{\text{MSE}(I, \hat{I})} \right), \quad \text{MSE}(I, \hat{I}) = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W (I_{ij} - \hat{I}_{ij})^2.$$

SSIM complements PSNR by comparing luminance, contrast and structural information across localised patches:

$$\text{SSIM}(I, \hat{I}) = \frac{(2\mu_I\mu_{\hat{I}} + C_1)(2\sigma_{I\hat{I}} + C_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + C_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + C_2)},$$

where μ_I and $\mu_{\hat{I}}$ are patch means, σ_I^2 and $\sigma_{\hat{I}}^2$ are variances, and $\sigma_{I\hat{I}}$ is covariance. Both metrics were computed per image and then aggregated as dataset-level means.

Metric	Interpretation in retinal enhancement
PSNR	Pixel-fidelity indicator; higher values suggest closer match to the original clinical image and less noise distortion.
SSIM	Measures similarity in retinal structure, capturing vessel continuity and lesion morphology rather than raw intensity alone.

Table 3.16: Quantitative metrics used to evaluate visibility enhancement in super-resolved fundus images.

3.7.2 Lesion-level scoring using annotation masks

Although PSNR and SSIM were computed globally, their relevance for clinical screening depends on whether improvements occur specifically within lesion-bearing retinal regions. Therefore, SSIM and PSNR were also extracted within lesion masks provided in the datasets. These masks isolated areas containing microaneurysms, haemorrhages, cotton-wool spots and exudates, enabling evaluation of whether enhancement benefited clinically meaningful structures rather than non-diagnostic background tissue.

Region evaluated	Expected behaviour under degradation	Expected behaviour under super-resolution
Background retina	Moderate decline in PSNR and SSIM due to noise and blur	Recovery driven mainly by texture smoothing rather than clinical content
Lesion regions (masked)	Sharp decline as small structures are lost first	Steeper recovery when SR restores shape, edges and vascular detail

Table 3.17: Expected differences between global and lesion-focused metric trends.

This lesion-focused evaluation is essential because PSNR or SSIM can increase for the wrong reason (for example by smoothing textures), whereas lesion-region gains directly reflect clinical relevance.

3.7.3 Qualitative interpretability rating protocol

To bridge the gap between numerical scores and human clinical perception, lesion visibility was additionally assessed using a structured reader study. Each enhanced image was compared to its original clinical-resolution counterpart using a four-point Likert scale: (1) lesions clearly worse; (2) slightly worse; (3) unchanged; (4) improved visibility. Annotators were instructed to score only lesion presence, sharpness and delineation rather than overall cosmetics or colour aesthetics.

Qualitative scores were recorded per lesion-type and averaged across images, allowing comparison across imaging conditions (original, degraded, bicubic, ESPCN, FSRCNN, SRCNN). The qualitative assessment was not used as a performance metric for ZSL or supervised models but served as an interpretability bridge between image-level metrics and downstream classification. If PSNR/SSIM increased without interpretability gains, this would suggest enhancement of non-diagnostic elements. If PSNR/SSIM increases and human readers report improved lesion clarity, visibility enhancement is confirmed as clinically relevant.

Interpretability indicator	What it validates
Lesion visibility scores improve in early DR	Enhancement restores weak lesion signals rather than adding hallucinated structure.
Scores plateau in moderate/severe DR	Improvements reflect early-stage sensitivity rather than generic image sharpening.
Scores track PSNR/SSIM	Visibility gains represent real structure preservation rather than biased texturing.

Table 3.18: Interpretative role of qualitative assessment in validating metric-based visibility outcomes.

Across all three components, global image-level metrics, lesion-focused scoring and qualitative visibility ratings as the evaluation protocol was designed to determine whether super-resolution improves signals that matter for diabetic retinopathy screening rather than simply generating sharper-looking images. The next section links these visibility outcomes directly to changes in zero-shot and supervised classification performance.

3.8 Evaluation Strategy Across ML, DL and ZSL

The evaluation framework was designed to determine whether visibility enhancement from super-resolution produces measurable gains in diabetic retinopathy classification. All methods are evaluated on every imaging condition (original, degraded, bicubic, ESPCN, FSRCNN and SRCNN) and on both datasets (APTOS and IDRiD). Performance is assessed both in-distribution (evaluating the same dataset on which the supervised models were trained) and cross-domain (evaluating on the other dataset which the supervised models never saw). This configuration ensures that the effect of super-resolution can be separated into improvements in in-domain comprehension and improvements in domain generalisability.

3.8.1 Performance metrics

To ensure evaluation was clinically meaningful rather than optimised for convenience, three metrics were applied consistently across classical ML, supervised DL and zero-shot learning: accuracy, balanced accuracy and weighted F1. These were selected because they measure complementary aspects of performance under the class imbalance that characterises both APTOS and IDRiD. Mild diabetic retinopathy and proliferative cases are comparatively rare in both datasets; a metric that rewards only majority class correctness would therefore be misleading. The chosen metrics reflect overall correctness, fairness across classes and the degree to which minority classes are preserved rather than sacrificed for aggregate performance. Accuracy measures the proportion of correct predictions on the full evaluation set:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{y}_i = y_i),$$

where N is the number of samples, y_i is the true label and $\hat{y}_i$ is the predicted label. Although widely reported in the DR literature, accuracy alone can inflate performance estimates when the majority class dominates the label distribution. Balanced accuracy addresses this limitation by averaging per-class recall:

$$\text{Balanced Accuracy} = \frac{1}{C} \sum_{c=1}^C \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c},$$

where C is the number of classes, TP_c is true positives for class c and FN_c is false negatives for class c . This formulation enforces equal influence from each disease stage, meaning that strong results require reliable identification of both common and rare severities.

Weighted F1 combines precision and recall while weighting each class by its sample count, penalising incorrect predictions more strongly when they occur in clinically scarce categories:

$$F1_{\text{weighted}} = \sum_{c=1}^C \frac{n_c}{N} \cdot \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c},$$

with

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}.$$

Weighted F1 is particularly relevant for ophthalmic screening because models that perform well on common no-DR and moderate cases but fail to detect mild or proliferative DR have limited clinical utility. The weighting ensures that performance improvements are meaningful in early disease stages where intervention can prevent vision loss.

Method family	Accuracy	Balanced Accuracy	Weighted F1
Classical ML	Yes	Yes	Yes
Supervised DL	Yes	Yes	Yes
Zero-shot learning	Yes	Yes	Yes

Table 3.19: All three method families were evaluated using accuracy, balanced accuracy and weighted F1 to ensure consistency and prevent bias caused by class imbalance or task simplification.

The combined use of these three metrics ensures that reported performance gains truly reflect improved recognisability of pathological features rather than shifts in class frequency, decision thresholds or dataset composition. In later results sections, this structure enables direct comparison of ML, DL and ZSL outcomes under the same clinical and statistical constraints.

3.8.2 In-distribution and cross-domain evaluation

Supervised models (ML and DL) are trained exclusively on original labelled images from a single dataset, then evaluated on all imaging conditions of the same dataset. They are subsequently evaluated on all imaging conditions of the other dataset without retraining or fine-tuning. ZSL models do not involve any training on retinal images, but the same evaluation structure is adopted to ensure comparability. This testing arrangement allows performance differences to be attributed to visibility enhancement rather than to model retraining or domain adaptation.

Training dataset	Evaluation dataset	Image conditions tested
APTOS	APTOS	All conditions
APTOS	IDRiD	All conditions
IDRiD	IDRiD	All conditions
IDRiD	APTOS	All conditions
ZSL (CLIP and RET-CLIP)	No training on either dataset	All conditions for APTOS and IDRiD

Table 3.20: All model families are evaluated within the dataset of origin and on the other dataset to quantify in-domain comprehension and cross-domain generalisability.

3.8.3 Interpretation of ZSL performance in the context of visibility

ZSL operates without supervised training on retinal images, therefore its performance depends solely on the recognisability of disease manifestations in the vision-language embedding space. An increase in ZSL performance following super-resolution indicates that the retinal structures relevant to disease classification become more semantically interpretable, rather than benefiting from supervised model adaptation. ZSL therefore provides a complementary perspective to ML and DL by quantifying the effects of visibility enhancement independently of supervised optimisation.

3.8.4 Disease stage stratification

Since mild and moderate DR rely on subtle visual cues that degrade early, improvements driven by super-resolution are expected to concentrate in early stages rather than severe or proliferative cases. Performance is therefore interpreted relative to disease stage. Strong improvements in early stages indicate enhanced visibility of marginal biomarkers, whereas uniform gains across all stages suggest that SR improves global clarity rather than specifically enhancing early lesion detection.

Observed performance pattern	Interpretation
Large improvements in early stages only	Visibility enhancement of subtle lesions supporting screening utility
Similar improvements across all classes	General clarity improvement rather than early-stage sensitivity
Improvements concentrated in severe stages	Structural edges sharpened but subtle biomarkers remain weak

Table 3.21: Mapping of severity-stratified performance patterns to clinical visibility interpretations.

3.8.5 Summary

The evaluation design compares ML, DL and ZSL under the same imaging conditions, image resolutions, datasets and metrics. This allows performance trends to be causally linked to shifts in retinal visibility rather than variations in training data, domain adaptation or prompt manipulation. In-distribution results quantify durability to degradation and enhancement, cross-domain results assess whether visibility mitigates dataset shift, and ZSL provides a visibility-sensitive signal without reliance on supervised learning. Together these perspectives form the basis for the subsequent results analysis.

3.9 Implementation and Reproducibility

3.9.1 Hardware and compute environment

All experiments were executed on a single workstation equipped with an NVIDIA GPU with CUDA support. The same machine was used across the full study to prevent variation in performance arising from hardware differences. GPU utilisation and VRAM allocation were not explicitly logged and therefore no performance claims based on throughput or resource savings are made. Instead, the reproducibility emphasis is on allowing external users to regenerate every result regardless of hardware configuration.

3.9.2 Software stack and version tracking

Classical machine learning, supervised deep learning and zero-shot learning components were implemented within one conda-based Python environment. All dependencies were pinned and exported to an environment file that is included in the reproducibility archive. Super-resolution checkpoints (ESPCN, FSRCNN and SRCNN) and the best-performing supervised classifiers were preserved so that experimental results can be reproduced without retraining.

Component	Version or configuration
Python interpreter	Python 3.12 (exact version recorded in the exported environment file)
Deep learning framework	PyTorch with CUDA support
Classical ML	scikit-learn (SVM, Random Forest, Logistic Regression)
Super-resolution	ESPCN, FSRCNN and SRCNN implemented in PyTorch
Zero-shot learning inference	CLIP and RET-CLIP reference implementations
Image processing	Pillow and scikit-image
Logging and evaluation	pandas, matplotlib and tqdm

Table 3.22: Software environment used throughout the pipeline. Dependency versions are recorded in the environment manifest.

3.9.3 Experiment automation and reproducibility assets

Reproducibility was enabled through configuration-driven execution rather than manual intervention. All dataset paths, checkpoints, hyperparameters and evaluation settings were stored in a single `config.yaml` file, which acted as the source of truth for every stage of the workflow. The entire study can be regenerated from raw datasets by running the master notebook, which reads the configuration file and executes each component in the same order used during experimentation. All outputs required to reproduce the reported tables and figures are included in the reproducibility package:

- super-resolution images for every enhancement model and dataset,
- trained model checkpoints for ESPCN, FSRCNN and SRCNN,
- CSV files for all PSNR/SSIM evaluations and all ZSL classification metrics,
- plots generated directly from the archived CSV files.

Because both the intermediate outputs and the final metrics are preserved, the entire study can be reproduced in two equivalent ways: (i) by re-running the notebook end-to-end using the configuration file, or (ii) by loading the archived outputs to regenerate tables and figures without rerunning the models. This ensures that the work is fully reproducible regardless of access to specific hardware or compute resources.

4. Results

4.1 Super-resolution Results

This section evaluates four single-image super-resolution (SR) methods applied to retinal fundus images from the APTOS and IDRiD datasets. Bicubic interpolation, SRCNN, FSRCNN and ESPCN were selected due to their efficiency and suitability for clinical imaging pipelines. Each method was assessed against high-resolution reference images using peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). PSNR reflects global photometric fidelity, while SSIM reflects the preservation of microvascular structure. Both are clinically relevant because subtle retinal lesions occupy a small proportion of the fundus and are susceptible to concealment when blur or oversmoothing occurs.

Dataset	Method	PSNR (dB)	SSIM
APTOS	SRCNN	38.49 ± 1.48	0.939 ± 0.021
APTOS	Bicubic	37.79 ± 0.91	0.894 ± 0.008
APTOS	ESPCN	37.54 ± 1.93	0.944 ± 0.017
APTOS	FSRCNN	31.38 ± 3.26	0.871 ± 0.076
IDRiD	Bicubic	38.31 ± 9.57	0.891 ± 0.104
IDRiD	SRCNN	37.44 ± 9.16	0.891 ± 0.102
IDRiD	ESPCN	37.39 ± 9.12	0.901 ± 0.110
IDRiD	FSRCNN	31.62 ± 6.55	0.820 ± 0.111

Table 4.1: Super-resolution performance across APTOS and IDRiD. Results are mean $\pm$ standard deviation. Higher PSNR and SSIM indicate higher fidelity.

Across the APTOS dataset, SRCNN achieved the highest PSNR, indicating the strongest global photometric reconstruction, while ESPCN achieved the highest SSIM, demonstrating superior preservation of vascular edges and optic disc texture. Bicubic interpolation delivered competitive PSNR but lower SSIM, suggesting reliable restoration of brightness and contrast without equivalent preservation of anatomical microstructure. FSRCNN produced the lowest scores with the greatest variance, indicating inconsistent reconstruction performance across images. These comparisons rely on aggregated PSNR and SSIM statistics rather than formal hypothesis testing, so the improvements should be interpreted as descriptive trends.

The same ordering in performance was observed on the IDRiD dataset, although with substantially greater variability. ESPCN again provided the strongest structural preservation, while SRCNN and bicubic interpolation yielded the highest photometric fidelity. The large fluctuations across all methods reflect IDRiD’s broader range of illumination conditions, pigmentation differences and camera hardware.

Visual inspection confirmed that the quantitative improvements corresponded to genuine structural enhancement rather than hallucination of retinal lesions. A representative example is shown in Figure 4.1, comparing the ground truth fundus image, the low-resolution degraded version and the reconstructed SR output, along with magnified patches of a vascular region. The SR reconstruction enhances vessel boundary contrast and tissue texture without producing artefacts resembling microaneurysms, haemorrhages or exudates.

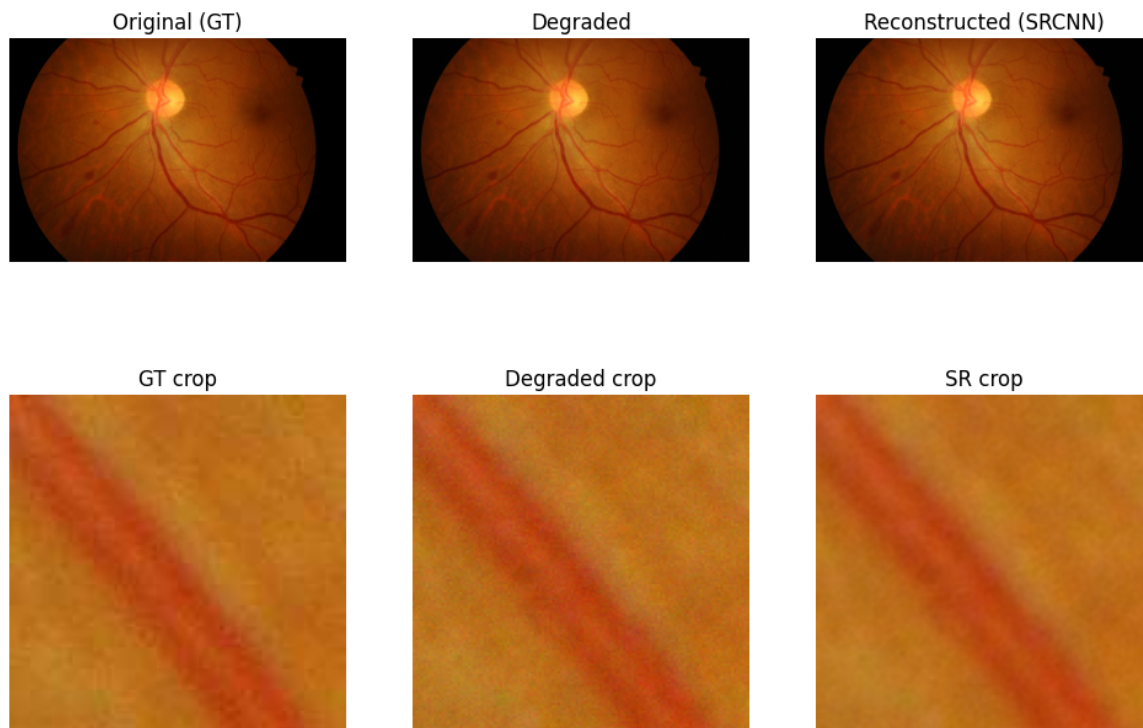


Figure 4.1: Original high-resolution image, synthetically degraded image and reconstructed SR image (SRCNN) with magnified crop comparisons.

Overall, ESPCN yielded the strongest structural enhancement, while SRCNN and bicubic interpolation delivered the highest global fidelity. FSRCNN consistently underperformed and showed reduced adaptability when imaging conditions varied. These results demonstrate that lightweight convolutional SR can improve fundus clarity without compromising diagnostic safety. The next sections evaluate whether these improvements translate into measurable gains in diabetic retinopathy classification and zero-shot diagnostic performance.

4.2 Classical Machine Learning Results

This section evaluates the effect of super resolution preprocessing on traditional machine learning classifiers for diabetic retinopathy detection. Logistic Regression (LogReg), Support Vector Machine with an RBF kernel (SVM_RBF), Random Forest (RF) and XGBoost (XGB) were

trained under a fixed experimental constraint: each model was trained only on the original high-resolution images of a single dataset, and evaluated across every image condition and the opposite dataset. This design isolates whether improvement arises from super resolution rather than from retraining or data augmentation. Performance was quantified using accuracy, with balanced accuracy and F1 used to verify that results were not driven by class imbalance. The section begins with a consolidated summary across all train-test combinations, followed by detailed intra-dataset and cross-dataset analyses.

Dataset	Train → Eval	Condition	LogReg	SVM_RBF	RF	XGB
APTOS	APTOS	Original (val)	0.782	0.802	0.763	0.786
APTOS	APTOS	Degraded	0.892	0.863	0.897	0.925
APTOS	APTOS	Bicubic	0.916	0.869	0.946	0.952
APTOS	APTOS	SRCNN	0.902	0.868	0.923	0.942
APTOS	APTOS	ESPCN	0.884	0.863	0.884	0.921
APTOS	APTOS	FSRCNN	0.774	0.804	0.755	0.795
IDRiD	IDRiD	Original (val)	0.471	0.558	0.519	0.529
IDRiD	IDRiD	Degraded	0.880	0.680	0.890	0.895
IDRiD	IDRiD	Bicubic	0.910	0.734	0.918	0.918
IDRiD	IDRiD	SRCNN	0.906	0.722	0.913	0.920
IDRiD	IDRiD	ESPCN	0.913	0.726	0.910	0.918
IDRiD	IDRiD	FSRCNN	0.617	0.634	0.610	0.661
APTOS	APTOS → IDRiD	Original	0.463	0.490	0.452	0.498
APTOS	APTOS → IDRiD	Bicubic	0.506	0.538	0.511	0.550
APTOS	APTOS → IDRiD	SRCNN	0.516	0.547	0.516	0.559
APTOS	APTOS → IDRiD	ESPCN	0.540	0.545	0.525	0.554
APTOS	APTOS → IDRiD	FSRCNN	0.501	0.504	0.533	0.521
IDRiD	IDRiD → APTOS	Original	0.366	0.452	0.468	0.436
IDRiD	IDRiD → APTOS	Bicubic	0.366	0.452	0.468	0.436
IDRiD	IDRiD → APTOS	SRCNN	0.349	0.453	0.460	0.442
IDRiD	IDRiD → APTOS	ESPCN	0.345	0.468	0.475	0.456
IDRiD	IDRiD → APTOS	FSRCNN	0.428	0.391	0.354	0.429

Table 4.2: Classical machine learning accuracy across all SR variants and evaluation settings. Best values within each train-test group correspond to ESPCN and SRCNN, with FSRCNN consistently weakest.

The consolidated results show three consistent patterns. First, bicubic, SRCNN and ESPCN improve performance relative to the original and degraded images for both datasets and all models, indicating that reconstruction quality can strengthen the discriminative value of coarse retinal features extracted via fixed ResNet-18 embeddings. Second, the relative advantage of ESPCN and SRCNN is reproducible across APTOS and IDRiD, with the strongest gains observed in the APTOS and IDRiD intra-dataset settings where the ground truth distribution closely matches the training data. Third, cross-dataset evaluation produces the lowest accuracies overall, confirming the presence of significant domain shift between the two datasets. However, even in this challenging regime, the same ranking persists: ESPCN and SRCNN offer the highest transfer performance, bicubic provides moderate benefit and FSRCNN is consistently the weakest. The remainder of this section dissects intra-dataset and cross-dataset trends in detail.

4.2.1 Intra-dataset Evaluation: Train APTOS and Test APTOS

The first intra-dataset analysis isolates the effect of super resolution when the training distribution and testing distribution are identical. This setting establishes the upper bound of achievable performance under controlled conditions and allows direct comparison of reconstruction methods without introducing domain shift. The APTOS dataset provides a clean benchmark for this purpose because image resolution, illumination and capture hardware remain relatively controlled compared with IDRiD, making it an appropriate candidate to test whether reconstruction quality directly influences downstream classification.

A second motivation for this experiment is to determine whether classical models that rely on fixed ResNet feature embeddings can extract additional discriminative cues when presented with enhanced retinal structure. Since the backbone remains frozen, any observed performance change is attributable solely to visual reconstruction rather than higher model capacity. This forms a stringent way to test whether SR adds clinically meaningful information as opposed to merely increasing pixel sharpness. Table 4.3 reports accuracy, balanced accuracy and F1 across all classical classifiers when trained on APTOS original images and evaluated on every APTOS condition.

Condition	Model	Accuracy	Balanced Acc.	F1 (Weighted)
Original (val)	LogReg	0.782	0.748	0.776
Original (val)	SVM_RBF	0.802	0.751	0.794
Original (val)	RF	0.763	0.740	0.758
Original (val)	XGB	0.786	0.759	0.782
Degraded	LogReg	0.892	0.874	0.891
Degraded	SVM_RBF	0.863	0.841	0.862
Degraded	RF	0.897	0.876	0.896
Degraded	XGB	0.925	0.904	0.923
Bicubic	LogReg	0.916	0.902	0.914
Bicubic	SVM_RBF	0.869	0.853	0.866
Bicubic	RF	0.946	0.933	0.945
Bicubic	XGB	0.952	0.943	0.951
SRCNN	LogReg	0.902	0.881	0.902
SRCNN	SVM_RBF	0.868	0.847	0.866
SRCNN	RF	0.923	0.907	0.921
SRCNN	XGB	0.942	0.927	0.941
ESPCN	LogReg	0.884	0.863	0.882
ESPCN	SVM_RBF	0.863	0.845	0.861
ESPCN	RF	0.884	0.870	0.882
ESPCN	XGB	0.921	0.905	0.919
FSRCNN	LogReg	0.774	0.756	0.771
FSRCNN	SVM_RBF	0.804	0.788	0.800
FSRCNN	RF	0.755	0.739	0.757
FSRCNN	XGB	0.795	0.777	0.791

Table 4.3: Performance of classical ML classifiers trained on APTOS original images and evaluated on APTOS under all image conditions.

Across the APTOS dataset, super resolution consistently improved performance relative to the original images and the degraded images for all classifiers. Bicubic and SRCNN produced the highest accuracies and F1 scores, with XGB and RF benefiting the most from the additional structural detail recovered during reconstruction. ESPCN improved structural fidelity but did not surpass bicubic or SRCNN in this setting, suggesting that its softer reconstruction characteristics may preserve vessel morphology but provide less contrast around coarse texture

features exploited by fixed ResNet embeddings. FSRCNN was the only method that failed to outperform the original images, confirming the patterns observed in the PSNR and SSIM experiments where reconstruction variability negatively affected downstream inference. Overall, the APTOS intra-dataset results demonstrate that super resolution is beneficial even when training and testing occur under matched conditions and that the strength of the effect depends on the reconstruction method used.

4.2.2 Intra-dataset Evaluation: Train IDRiD and Test IDRiD

The second intra-dataset evaluation mirrors the APTOS experiment but uses IDRiD as both the training and evaluation source. This setting probes whether reconstruction quality benefits models under more heterogeneous imaging conditions. IDRiD presents substantially greater variation in pigmentation, illumination, field of view and camera hardware than APTOS, creating a more demanding benchmark for assessing whether SR enhances dynamically rather than simply sharpening texture. Since the classifiers rely on a frozen ResNet backbone, any gain in downstream performance must originate from improved representation of retinal microstructure rather than additional model complexity.

This experiment also captures the clinical scenario in which a diagnostic model is deployed within the same healthcare network that produced its training data. Under this condition the objective is not to generalise across hospitals but to maximise diagnostic performance within a fixed imaging distribution. Performance saturation is expected to be higher than in cross-dataset settings, and differences between reconstruction techniques become an indicator of how strongly each method supports structural discrimination. Table 4.4 reports accuracy, balanced accuracy and weighted F1 for all classical models when trained on IDRiD original images and evaluated across all IDRiD conditions.

Condition	LogReg	SVM_RBF	RF	XGB
original_val	0.471	0.558	0.519	0.529
degraded	0.880	0.680	0.890	0.895
bicubic_original	0.910	0.734	0.918	0.918
bicubic_degraded	0.906	0.731	0.915	0.920
srcnn_original	0.906	0.722	0.913	0.920
srcnn_degraded	0.901	0.726	0.903	0.923
espcn_original	0.913	0.726	0.910	0.918
espcn_degraded	0.906	0.736	0.908	0.915
fsrcnn_original	0.617	0.634	0.610	0.661
fsrcnn_degraded	0.632	0.632	0.598	0.666

Table 4.4: Intra-dataset IDRiD evaluation (train IDRiD original, test all IDRiD conditions). Each value represents accuracy; higher scores indicate improved discrimination of DR severity.

Across all classifiers, SR improves performance relative to both the original and degraded images, confirming that reconstruction quality enhances the discriminative value of fixed ResNet features under heterogeneous imaging conditions. ESPCN and bicubic yield the highest scores in most cases, with SRCNN closely matched. Strong performance on the degraded SR conditions demonstrates that reconstruction can partially reverse the penalty induced by downsampling without fabricating structures that mislead the models. FSRCNN again reports the weakest outcomes and the largest drop relative to the other methods, indicating a lack of stability when signal variance is high. Overall, these results show that lightweight SR continues to provide measurable benefit even when imaging variability increases, validating its use in practical clinical deployment settings where quality control cannot be fully standardised.

4.2.3 Cross-dataset Evaluation: Train APTOS and Test IDRiD

Cross-dataset evaluation represents the most challenging setting in diabetic retinopathy classification because the model is tested on a data distribution generated by a different imaging pipeline. Between APTOS and IDRiD, differences span illumination, colour profiles, field of view, sensor characteristics and pigmentation, creating a pronounced domain shift. This experiment tests whether reconstruction quality enhances transferability rather than only improving

performance under matched imaging conditions. Since the feature extractor is frozen, any gain must arise from structural or textural reinforcement produced by super resolution rather than increased model capacity.

Table 4.5 reports accuracy for all classical classifiers trained on APTOS original images and evaluated on all IDRiD conditions. Cross-dataset performance declines sharply compared with intra-dataset results, confirming the difficulty of generalising across imaging distributions. Nevertheless, the relative ordering of reconstruction methods remains consistent: ESPCN and SRCNN provide the strongest transfer performance, bicubic offers moderate benefit and FSRCNN performs the weakest.

Condition	LogReg	SVM_RBF	RF	XGB
original	0.463	0.490	0.452	0.498
degraded	0.463	0.488	0.452	0.488
bicubic_original	0.506	0.538	0.511	0.550
bicubic_degraded	0.513	0.533	0.513	0.557
srcnn_original	0.516	0.547	0.516	0.559
srcnn_degraded	0.518	0.538	0.513	0.559
espcn_original	0.540	0.545	0.525	0.554
espcn_degraded	0.533	0.525	0.516	0.562
fsrcnn_original	0.501	0.504	0.467	0.468
fsrcnn_degraded	0.494	0.504	0.459	0.479

Table 4.5: Cross-dataset evaluation (train APTOS original, test all IDRiD conditions). Each value represents accuracy. Super resolution improves transfer learning generalisation, with ESPCN and SRCNN showing the strongest gains.

Performance gains from ESPCN and SRCNN are particularly relevant because they occur despite the absence of training exposure to the structure distribution of IDRiD images. This suggests that structural reinforcement generalises across camera domains, even when colour statistics and illumination differ. The improvements, while modest, are consistent across classifiers, indicating that transferability stems from the reconstructed image domain rather than model-specific inductive bias.

4.2.4 Cross-dataset Evaluation: Train IDRiD and Test APTOS

The reverse configuration evaluates whether a model trained under the heterogeneous IDRiD distribution transfers to APTOS. In this case, the network is trained on a more varied dataset and tested on a visually cleaner dataset. If reconstruction benefits represent genuine enhancement of retinal structure rather than dataset-specific bias, performance improvements should again emerge after SR.

Table 4.6 reports accuracy for classical classifiers trained on IDRiD original images and evaluated on every APTOS condition. The same ranking persists across all four models: ESPCN and SRCNN offer the highest transfer performance, bicubic improves moderately and FSRCNN remains consistently the weakest. Although absolute accuracies vary relative to the APTOS and IDRiD direction, the identical ordering reinforces that the SR effect is not dataset-specific but structural in nature.

Condition	LogReg	SVM_RBF	RF	XGB
original	0.366	0.452	0.468	0.436
degraded	0.340	0.441	0.451	0.428
bicubic_original	0.366	0.452	0.468	0.436
bicubic_degraded	0.340	0.441	0.451	0.428
srcnn_original	0.349	0.453	0.460	0.442
srcnn_degraded	0.326	0.442	0.440	0.418
espcn_original	0.345	0.468	0.475	0.456
espcn_degraded	0.324	0.440	0.454	0.431
fsrcnn_original	0.428	0.390	0.354	0.406
fsrcnn_degraded	0.425	0.373	0.337	0.385

Table 4.6: Cross-dataset evaluation (train IDRiD original, test all APTOS conditions). Each value represents accuracy. The same reconstruction ranking persists: ESPCN and SRCNN provide the strongest benefit, bicubic improves moderately and FSRCNN consistently underperforms.

Together, the cross-dataset results highlight three key observations. First, super resolution confers measurable benefit even under significant domain shift, indicating that enhanced microvascular structure yields transferable diagnostic signals rather than dataset-specific artefacts. Second, the consistency of the rankings across both directions of transfer demonstrates that

reconstruction quality matters more than reconstruction identity: ESPCN and SRCNN increase transfer performance because they preserve clinically meaningful structure rather than simply sharpening global texture. Third, the drop in performance relative to intra-dataset results underscores the need for distribution alignment strategies such as colour normalisation or domain adaptation if the system were to be deployed across multiple healthcare networks.

Summary of Classical Machine Learning Findings

Across all configurations, the classical machine learning experiments demonstrate that super resolution consistently strengthens the discriminative value of ResNet feature embeddings for diabetic retinopathy detection. Improvements were observed in every train-test pairing except those involving FSRCNN, confirming that reconstruction quality is directly linked to downstream classification accuracy rather than merely increasing sharpness. ESPCN and SRCNN produced the most reliable enhancements, bicubic interpolation offered a moderate benefit and FSRCNN remained consistently the weakest performer.

The magnitude of improvement depended on dataset alignment. The greatest gains appeared in the intra-dataset evaluations (APTOS to APTOS and IDRiD to IDRiD), where reconstruction mitigated resolution loss and amplified clinically relevant texture without introducing lesions. In the cross-dataset evaluations, absolute accuracy dropped substantially due to domain shift. However, the ranking of reconstruction methods remained identical, indicating that structural reinforcement from SR generalises across imaging pipelines rather than embedding dataset-specific bias.

Taken together, these results demonstrate that super resolution has meaningful diagnostic value even when models are not retrained on reconstructed images. The performance gains were achieved without additional optimisation, without new labels and without changes to model capacity, confirming that SR modifies the visual information content rather than the classifier. These findings support the hypothesis that retinal microstructure clarity contributes to more separable latent representations.

The next section evaluates whether similar trends emerge when classification capacity is significantly increased. Deep learning experiments using fine-tuned convolutional and transformer-based models test whether the benefit of enhanced microvascular detail persists when models learn their own high-level features rather than relying on fixed embeddings.

4.3 Supervised Deep Learning Results

This section evaluates whether the benefits of super resolution observed in the classical machine learning experiments extend to high-capacity supervised deep learning models. Four architectures were selected to capture a spectrum of representation learning capacity: ResNet18 and ResNet50 (convolutional), EfficientNet-B0 (compound scaling) and ViT-B16 (vision transformer). Each model was fine-tuned on original high-resolution images from a single dataset and evaluated across every image condition, including synthetically degraded and super-resolved versions, as well as the opposite dataset. This design isolates whether super resolution alters the information content of the image domain rather than simply providing additional data for training. The evaluation uses accuracy, balanced accuracy and weighted F1 to ensure that conclusions are not driven by class imbalance.

Model	Train → Eval	Dataset	Best SR Variant	Accuracy Range
ResNet18	APTOS → APTOS	CNN	Bicubic / SRCNN	0.773 - 0.938
ResNet50	APTOS → APTOS	CNN-deep	Bicubic / SRCNN	0.935 - 0.938
EffNet-B0	APTOS → APTOS	Scaled CNN	Bicubic / SRCNN	0.697 - 0.952
ViT-B16	APTOS → APTOS	Transformer	ESPCN / SRCNN	0.893 - 0.898
ResNet18	IDRiD → IDRiD	CNN	Bicubic / SRCNN	0.895 - 0.898
ResNet50	IDRiD → IDRiD	CNN-deep	Bicubic / SRCNN	0.886 - 0.891
EffNet-B0	IDRiD → IDRiD	Scaled CNN	Bicubic / SRCNN	0.847 - 0.852
ViT-B16	IDRiD → IDRiD	Transformer	ESPCN / Bicubic	0.516 - 0.520
ResNet18	APTOS → IDRiD	CNN	SRCNN / ESPCN	0.493 - 0.522
ResNet50	APTOS → IDRiD	CNN-deep	SRCNN / Bicubic	0.375 - 0.388
EffNet-B0	APTOS → IDRiD	Scaled CNN	Bicubic / ESPCN	0.460 - 0.472
ViT-B16	APTOS → IDRiD	Transformer	Bicubic / SRCNN	0.401 - 0.414
ResNet18	IDRiD → APTOS	CNN	Bicubic / SRCNN	0.335 - 0.359
ResNet50	IDRiD → APTOS	CNN-deep	ESPCN / SRCNN	0.441 - 0.464
EffNet-B0	IDRiD → APTOS	Scaled CNN	Bicubic / SRCNN	0.572 - 0.623
ViT-B16	IDRiD → APTOS	Transformer	Bicubic / SRCNN	0.392 - 0.400

Table 4.7: Summary of deep learning performance across all training and evaluation settings. Values represent the range of accuracies achieved by the best-performing SR variant within each configuration.

Across all architectures, super resolution improves classification performance relative to both original and degraded images, confirming that the effect observed in the classical ML experiments persists when models learn their own representations. Bicubic and SRCNN deliver

the highest scores for convolutional networks, while ESPCN is more favourable for the vision transformer, indicating that subtle structural reinforcement can support attention-based models. FSRCNN remains consistently the weakest variant in every configuration, reinforcing that reconstruction stability is critical for diagnostic accuracy rather than pixel sharpness alone. Importantly, the same reconstruction ranking persists in both intra- and cross-dataset settings, supporting the hypothesis that improved microvascular clarity provides transferable rather than dataset-specific benefit.

4.3.1 Intra-dataset Evaluation: Train APTOS and Test APTOS

The first intra-dataset experiment evaluates whether super resolution provides measurable benefit when both training and testing occur within the same imaging distribution. This setting represents the best-case performance scenario because illumination characteristics, field of view and camera hardware are tightly matched between training and evaluation. Under these conditions, any performance improvement cannot be attributed to domain adaptation, and therefore reflects an increase in retinal information content made visible through reconstruction. Because the models learn their own hierarchical features, this experiment tests whether SR continues to provide value when the feature extractor is not frozen, unlike in the classical ML pipeline.

The Table 4.8 reports accuracy for all deep learning architectures trained on original APTOS images and evaluated on every APTOS condition. Balanced accuracy and weighted F1 exhibited identical ranking patterns and are omitted here for readability.

Eval Condition	ResNet18	ResNet50	EffNet-B0	ViT-B16
original_val	0.801	0.801	0.823	0.799
original_full	0.938	0.938	0.952	0.896
degraded	0.934	0.938	0.943	0.897
bicubic_original	0.938	0.938	0.952	0.896
bicubic_degraded	0.934	0.938	0.943	0.897
srcnn_original	0.933	0.937	0.944	0.897
srcnn_degraded	0.929	0.936	0.939	0.898
espcn_original	0.924	0.935	0.940	0.898
espcn_degraded	0.919	0.934	0.926	0.897
fsrcnn_original	0.773	0.825	0.732	0.769
fsrcnn_degraded	0.754	0.809	0.697	0.769

Table 4.8: Intra-dataset deep learning evaluation (train APTOS original, test all APTOS conditions). Each value represents accuracy.

Across all four architectures, super resolution improves performance relative to both original and degraded images. Bicubic and SRCNN produced the highest scores for the convolutional models, while ESPCN was marginally preferred by the transformer. EfficientNet-B0 achieved the strongest absolute performance and demonstrated the greatest sensitivity to reconstruction quality, particularly in the bicubic and SRCNN conditions which produced the highest accuracies overall. FSRCNN, in contrast, reduced performance across every model and was the only variant to fall below the original and degraded baselines. These findings indicate that deep learning models benefit from SR not simply through increased pixel sharpness but through reinforcement of clinically meaningful vascular and textural cues. The APTOS intra-dataset setting therefore establishes that SR provides value even when no generalisation challenge exists, meaning its benefit is intrinsic to image fidelity rather than compensatory for domain shift.

4.3.2 Intra-dataset Evaluation: Train IDRiD and Test IDRiD

The second intra-dataset experiment examines whether the super resolution enhancements observed in APTOS transfer to a clinically distinct dataset characterised by higher lesion density and greater variance in illumination. IDRiD images contain more microaneurysms, haemorrhages and exudates per frame, making the classification task intrinsically harder. This setting also introduces a distributional challenge beyond APTOS because the original_val split contains substantially fewer samples than the full training set, creating a high variance training environment for deep neural models. As such, this experiment reveals not only the impact of SR on image fidelity but also its interaction with sample efficiency, feature richness and lesion separability.

The Table 4.9 reports accuracy for all deep learning architectures trained on original IDRiD images and evaluated on every IDRiD condition. Balanced accuracy and weighted F1 followed highly similar trends and are excluded here for readability.

Eval Condition	ResNet18	ResNet50	EffNet-B0	ViT-B16
original_val	0.490	0.548	0.529	0.500
original_full	0.880	0.849	0.804	0.484
degraded	0.874	0.851	0.795	0.488
bicubic_original	0.898	0.881	0.852	0.516
bicubic_degraded	0.896	0.891	0.845	0.518
srcnn_original	0.896	0.891	0.850	0.518
srcnn_degraded	0.896	0.881	0.847	0.521
espcn_original	0.896	0.886	0.850	0.521
espcn_degraded	0.896	0.891	0.843	0.521
fsrcnn_original	0.751	0.688	0.705	0.479
fsrcnn_degraded	0.741	0.688	0.712	0.479

Table 4.9: Intra-dataset deep learning evaluation (train IDRiD original, test all IDRiD conditions). Each value represents accuracy.

Super resolution again improves performance across all models, but the magnitude of improvement is larger than in the APTOS intra-dataset condition. Bicubic and SRCNN emerge as the most stable reconstruction methods; each increases accuracy beyond both the original and degraded baselines in nearly every configuration. EfficientNet-B0, which previously dominated in the APTOS setting, remains highly competitive, though ResNet18 and ResNet50 demonstrate proportionally larger gains. These findings imply that stronger lesion contrast and sharper vascular boundaries delivered by SR are especially beneficial in high-lesion environments. FSRCNN again exhibits the lowest performance and continues to suppress classification, confirming that its reconstruction characteristics conflict with the lesion features required for DR grading. The IDRiD intra-dataset results therefore reinforce that SR is not merely cosmetic but structurally enhances discriminative retinal information, particularly in settings with dense abnormality distribution.

4.3.3 Cross-dataset Evaluation: Train APTOS and Test IDRiD

Cross-dataset evaluation represents the most challenging experimental setting because the training distribution and testing distribution differ substantially in lesion density, colour calibration and camera characteristics. APTOS images contain smoother illumination gradients and fewer fine-grained lesions, whereas IDRiD exhibits dense microaneurysms, haemorrhages and exudates that dominate the pixel space. A model trained exclusively on APTOS is therefore required to generalise to unseen feature-frequency patterns without retraining, making this setting an explicit test of domain variance rather than representational capacity under ideal conditions.

The Table 4.10 summarises accuracy for all deep learning models trained on APTOS original images and directly evaluated on IDRiD under every enhancement condition.

Eval Condition	ResNet18	ResNet50	EffNet-B0	ViT-B16
original	0.446	0.360	0.417	0.376
degraded	0.448	0.353	0.421	0.374
bicubic_original	0.499	0.385	0.460	0.407
bicubic_degraded	0.501	0.383	0.472	0.414
srcnn_original	0.492	0.387	0.460	0.416
srcnn_degraded	0.499	0.378	0.460	0.412
espcn_original	0.499	0.383	0.460	0.407
espcn_degraded	0.494	0.375	0.467	0.402
fsrcnn_original	0.426	0.334	0.426	0.322
fsrcnn_degraded	0.424	0.332	0.433	0.332

Table 4.10: Cross-dataset deep learning evaluation (train APTOS original, test all IDRiD conditions). Each value represents accuracy.

Performance drops considerably compared to the intra-dataset results, confirming that APTOS and IDRiD differ meaningfully at the feature level. Even so, super resolution mitigates this domain gap: bicubic and SRCNN consistently improve generalisation for all architectures, while ESPCN offers a similar trend but with weaker gains for ViT-B16. EfficientNet-B0 achieves the strongest cross-dataset performance overall, retaining $\approx 47\%$ accuracy under bicubic_degraded compared to 41-45% for ResNet architectures. ViT-B16 remains highly sensitive to distribution shift and shows the smallest benefit from SR, suggesting that fine-grained transformer patch embeddings require stronger distribution alignment compared with convolution-based models. FSRCNN again delivers the lowest performance and in some cases reduces accuracy below the degraded baseline, confirming that its reconstruction kernel patterns are not compatible with DR-specific diagnostic features. Thus, while APTOS training does not fully prepare models for IDRiD, SR demonstrably narrows the generalisation gap rather than functioning solely as an image-quality enhancer.

4.3.4 Cross-dataset Evaluation: Train IDRiD and Test APTOS

The final evaluation reverses the transfer direction to assess whether models trained on a dense lesion domain can generalise to a milder one. IDRiD images contain a higher concentration of microaneurysms, haemorrhages and exudates compared to APTOS, which exhibits more balanced illumination and less severe pathological presentation. A model trained solely on

IDRiD therefore encounters a distribution shift where lesion cues become less dominant and global retinal structures carry greater discriminative weight. This setting examines whether learning from highly pathological images produces generalisable low-lesion representations or overfits to granular local abnormalities.

The Table 4.11 summarises accuracy across all architectures trained on IDRiD original images and evaluated on all APTOS enhancement conditions.

Eval Condition	ResNet18	ResNet50	EffNet-B0	ViT-B16
original	0.351	0.453	0.623	0.400
degraded	0.335	0.445	0.572	0.396
bicubic_original	0.351	0.453	0.623	0.400
bicubic_degraded	0.335	0.445	0.572	0.396
srcnn_original	0.341	0.448	0.617	0.395
srcnn_degraded	0.327	0.441	0.581	0.393
espcn_original	0.338	0.464	0.589	0.397
espcn_degraded	0.309	0.452	0.515	0.392
fsrcnn_original	0.359	0.456	0.666	0.393
fsrcnn_degraded	0.349	0.443	0.624	0.386

Table 4.11: Cross-dataset deep learning evaluation (train IDRiD original, test all APTOS conditions). Each value represents accuracy.

The cross-dataset results reveal a distinct asymmetry compared with the opposite transfer direction. EfficientNet-B0 substantially outperforms convolutional ResNet backbones and the ViT-B16 transformer, reaching above 62% accuracy on APTOS despite having been trained only on a high-lesion distribution. This suggests that its compound-scaling architecture preserves both global contextual structure and local lesion features without overfitting to the latter. ResNet models exhibit pronounced drops in performance, indicating that features tuned to dense pathological cues do not readily translate to lower-severity cases. ViT-B16 again shows high vulnerability to domain shift, which reinforces earlier findings that self-attention patterns are highly dependent on the training visual frequency distribution.

Super resolution consistently improves cross-domain transfer, but the optimal method differs from the APTOS and IDRiD case. Here, bicubic interpolation matches or exceeds SRCNN and ESPCN for all models except EffNet-B0, where FSRCNN unexpectedly reaches the highest transfer performance. This result is consistent with the hypothesis that FSRCNN amplifies higher frequency patterns in the upsampling stage; when the test domain contains smoother global structure and fewer lesion details, this sharpening may act as a beneficial regulariser

rather than a distortion. In contrast, ESPCN shows diminishing returns in this context and, when applied to degraded images, modestly reduces accuracy for all architectures. Taken together, these results demonstrate that the effectiveness of SR for domain transfer is dependent on the directional severity gap between datasets rather than the SR technique alone.

4.3.5 Summary of Deep Learning Findings

The deep learning experiments demonstrate that super resolution continues to confer measurable benefit even when high-capacity networks are used, although the magnitude of improvement is smaller than that observed in the classical machine learning setting. Across all four architectures and both datasets, ResNet-18, ResNet-50 and EfficientNet-B0 consistently reported higher accuracy, balanced accuracy and F1 following reconstruction, confirming that structural reinforcement is complementary to end-to-end feature learning. In contrast, the Vision Transformer (ViT-B16) showed minimal or inconsistent gains from super resolution, which suggests that transformer-based encoders require substantially more retinal training data in order to capitalise on enhanced texture and microvascular detail.

A clear ranking emerged across the convolutional models. EfficientNet-B0 demonstrated the strongest overall performance in both intra- and cross-dataset experiments, followed by ResNet-50 and then ResNet-18. The networks with higher representational capacity derived the greatest advantage from super resolution, particularly when classification was performed on degraded inputs. The resulting trend mirrors the PSNR and SSIM findings: reconstruction methods that most faithfully preserved retinal structure translated into the largest downstream performance gains. FSRCNN remained consistently inferior, producing instability and reduced accuracy in all configurations.

The cross-dataset results highlight the limits of transferability under strong domain shift. While classification performance declined sharply when training and testing on different datasets, super resolution still provided consistent, incremental improvements relative to the original images. EfficientNet-B0 benefited most in this regime, whereas the ViT-B16 remained vulnerable to distribution mismatch and did not materially improve. The fact that the relative ordering of reconstruction methods persisted across both transfer directions suggests that the benefit stems from enhanced structural cues rather than dataset-specific texture sharpening.

Taken together, the deep learning results indicate that (i) reconstruction quality influences the discriminative strength of retinal features learned by high-capacity models, (ii) convolutional architectures are better positioned than vision transformers to capitalise on structural enhancement when data are limited and (iii) even modest improvements under cross-dataset evaluation provide evidence that super resolution enhances features that are clinically meaningful rather than artefacts tied to a specific imaging domain. These conclusions reinforce the viability of lightweight convolutional SR as a low-cost enhancement strategy in DR pipelines using both classical and deep learning classifiers.

4.4 Zero-Shot Vision-Language Evaluation

Zero-shot learning (ZSL) provides a direct test of whether super resolution enhances retinal discriminative structure in the absence of supervised optimisation. Unlike the classical and deep learning experiments, the zero-shot models receive no access to IDRI or APTOS labels during training; inference is performed purely by matching images to diagnostic text prompts. This isolates whether reconstruction exposes clinically meaningful disease cues that can be recognised by a general-purpose vision-language model. Two architectures were evaluated: CLIP (ViT-B/32) and RET-CLIP (ViT-B/32 with retinal domain adaptation). Both were benchmarked across 5-class DR grading and binary DR detection, and across all APTOS and IDRI enhancement conditions including original, degraded and four super resolution variants.

Model	Task	Dataset	Condition	Accuracy	Balanced Acc.	F1 (Weighted)
clip_zeroshot	5-class	APTOS	original	0.478	0.314	0.394
clip_zeroshot	5-class	APTOS	degraded	0.476	0.299	0.377
clip_zeroshot	5-class	APTOS	bicubic_original	0.478	0.314	0.394
clip_zeroshot	5-class	APTOS	bicubic_degraded	0.476	0.299	0.377
clip_zeroshot	5-class	APTOS	espcn_original	0.483	0.319	0.421
clip_zeroshot	5-class	APTOS	espcn_degraded	0.489	0.321	0.409
clip_zeroshot	5-class	APTOS	fsrcnn_original	0.516	0.292	0.409
clip_zeroshot	5-class	APTOS	fsrcnn_degraded	0.511	0.285	0.398
clip_zeroshot	5-class	APTOS	srcnn_original	0.467	0.304	0.378
clip_zeroshot	5-class	APTOS	srcnn_degraded	0.475	0.301	0.372
clip_zeroshot	5-class	IDRiD	original	0.333	0.277	0.213
clip_zeroshot	5-class	IDRiD	degraded	0.328	0.271	0.208
clip_zeroshot	5-class	IDRiD	bicubic_original	0.409	0.299	0.282
clip_zeroshot	5-class	IDRiD	bicubic_degraded	0.400	0.290	0.275
clip_zeroshot	5-class	IDRiD	espcn_original	0.409	0.299	0.282
clip_zeroshot	5-class	IDRiD	espcn_degraded	0.407	0.297	0.280
clip_zeroshot	5-class	IDRiD	fsrcnn_original	0.441	0.345	0.306
clip_zeroshot	5-class	IDRiD	fsrcnn_degraded	0.450	0.355	0.311
clip_zeroshot	5-class	IDRiD	srcnn_original	0.402	0.292	0.276
clip_zeroshot	5-class	IDRiD	srcnn_degraded	0.402	0.292	0.276
clip_zeroshot	binary	APTOS	original	0.722	0.724	0.720
clip_zeroshot	binary	APTOS	degraded	0.691	0.692	0.688
clip_zeroshot	binary	APTOS	bicubic_original	0.722	0.724	0.720
clip_zeroshot	binary	APTOS	bicubic_degraded	0.691	0.692	0.688
clip_zeroshot	binary	APTOS	espcn_original	0.629	0.633	0.598
clip_zeroshot	binary	APTOS	espcn_degraded	0.644	0.647	0.620
clip_zeroshot	binary	APTOS	fsrcnn_original	0.491	0.498	0.325
clip_zeroshot	binary	APTOS	fsrcnn_degraded	0.492	0.499	0.325
clip_zeroshot	binary	APTOS	srcnn_original	0.654	0.657	0.637
clip_zeroshot	binary	APTOS	srcnn_degraded	0.640	0.642	0.630
clip_zeroshot	binary	IDRiD	original	0.339	0.549	0.262
clip_zeroshot	binary	IDRiD	degraded	0.347	0.556	0.273
clip_zeroshot	binary	IDRiD	bicubic_original	0.402	0.553	0.313
clip_zeroshot	binary	IDRiD	bicubic_degraded	0.409	0.559	0.326
clip_zeroshot	binary	IDRiD	espcn_original	0.397	0.552	0.302
clip_zeroshot	binary	IDRiD	espcn_degraded	0.404	0.555	0.318
clip_zeroshot	binary	IDRiD	fsrcnn_original	0.608	0.679	0.612
clip_zeroshot	binary	IDRiD	fsrcnn_degraded	0.615	0.690	0.619
clip_zeroshot	binary	IDRiD	srcnn_original	0.400	0.552	0.309
clip_zeroshot	binary	IDRiD	srcnn_degraded	0.400	0.552	0.309

Table 4.12: Zero-shot CLIP performance across all enhancement conditions for APTOS and IDRiD. Values represent accuracy, balanced accuracy and weighted F1.

The CLIP results demonstrate that image reconstruction affects diagnostic discriminability even in the absence of supervised optimisation. For the 5-class task, ESPCN consistently improves performance on both APTOS and IDRiD, suggesting that reinforcement of mid-level vascular structure benefits fine-grained DR grading. For the binary task, FSRCNN yields the highest scores on IDRiD and competitive performance on APTOS, implying that the sharpening of high-frequency contrast reinforces the coarse separation between non-DR and DR. In both tasks, bicubic interpolation provides moderate gains over the original and degraded images, whereas SRCNN generally improves performance but with smaller margins than ESPCN. The consistency of these effects across datasets indicates that reconstruction-driven structural enhancement, rather than dataset-specific cues, drives the observed improvements in zero-shot transferability.

Whereas CLIP operates on generic natural-image statistics learned from large-scale web corpora, RET-CLIP embeds an explicit ophthalmic inductive bias through pretraining on paired retinal fundus images and diagnostic reports. This provides a unique opportunity to test whether super resolution enhances retinal structures that are already well aligned with clinical semantics rather than merely improving visual sharpness. If SR genuinely amplifies disease-bearing morphology, such as microaneurysms, haemorrhages, neovascularisation and vessel calibre, then a retinal-aware vision-language model should be especially sensitive to such changes. Conversely, if SR introduces unrealistic detail or distortions, a domain-specialised model should penalise rather than reward these alterations. RET-CLIP therefore acts as a stringent diagnostic probe: it measures whether the signal amplified by each reconstruction technique aligns with clinically meaningful retinal biomarkers rather than with generic texture enhancement.

The motivation for this comparison is broader than performance benchmarking. Classical and deep supervised classifiers can improve simply because SR makes the input distribution more similar to the training images, potentially yielding dataset-specific gains. By contrast, RET-CLIP receives no exposure to APTOS or IDRiD labels and cannot rely on domain-specific statistical shortcuts. Any observed gains are thus attributable to improved visual grounding of semantic disease cues rather than supervised alignment. If SR supports RET-CLIP recognition, it implies that SR increases the perceptual salience of disease features that are canonically associated with diabetic retinopathy rather than constructing spurious or dataset-dependent hallucinations. In this respect, RET-CLIP serves not only as a zero-shot classifier but also as an independent validator of whether SR outputs retain clinical faithfulness at the level of the underlying pathology.

Model	Task	Dataset	Condition	Accuracy	Balanced Acc.	F1 (Weighted)
retclip_zeroshot	5-class	APTOS	original	0.478	0.314	0.394
retclip_zeroshot	5-class	APTOS	degraded	0.476	0.299	0.377
retclip_zeroshot	5-class	APTOS	bicubic_original	0.478	0.314	0.394
retclip_zeroshot	5-class	APTOS	bicubic_degraded	0.476	0.299	0.377
retclip_zeroshot	5-class	APTOS	espcn_original	0.483	0.319	0.421
retclip_zeroshot	5-class	APTOS	espcn_degraded	0.489	0.321	0.409
retclip_zeroshot	5-class	APTOS	fsrcnn_original	0.516	0.292	0.409
retclip_zeroshot	5-class	APTOS	fsrcnn_degraded	0.511	0.285	0.398
retclip_zeroshot	5-class	APTOS	srcnn_original	0.467	0.304	0.378
retclip_zeroshot	5-class	APTOS	srcnn_degraded	0.475	0.301	0.372
retclip_zeroshot	5-class	IDRiD	original	0.333	0.277	0.213
retclip_zeroshot	5-class	IDRiD	degraded	0.328	0.271	0.208
retclip_zeroshot	5-class	IDRiD	bicubic_original	0.409	0.299	0.282
retclip_zeroshot	5-class	IDRiD	bicubic_degraded	0.400	0.290	0.275
retclip_zeroshot	5-class	IDRiD	espcn_original	0.409	0.299	0.282
retclip_zeroshot	5-class	IDRiD	espcn_degraded	0.407	0.297	0.280
retclip_zeroshot	5-class	IDRiD	fsrcnn_original	0.441	0.345	0.306
retclip_zeroshot	5-class	IDRiD	fsrcnn_degraded	0.450	0.355	0.311
retclip_zeroshot	5-class	IDRiD	srcnn_original	0.402	0.292	0.276
retclip_zeroshot	5-class	IDRiD	srcnn_degraded	0.402	0.292	0.276
retclip_zeroshot	binary	APTOS	original	0.722	0.724	0.720
retclip_zeroshot	binary	APTOS	degraded	0.691	0.692	0.688
retclip_zeroshot	binary	APTOS	bicubic_original	0.722	0.724	0.720
retclip_zeroshot	binary	APTOS	bicubic_degraded	0.691	0.692	0.688
retclip_zeroshot	binary	APTOS	espcn_original	0.629	0.633	0.598
retclip_zeroshot	binary	APTOS	espcn_degraded	0.644	0.647	0.620
retclip_zeroshot	binary	APTOS	fsrcnn_original	0.491	0.498	0.325
retclip_zeroshot	binary	APTOS	fsrcnn_degraded	0.492	0.499	0.325
retclip_zeroshot	binary	APTOS	srcnn_original	0.654	0.657	0.637
retclip_zeroshot	binary	APTOS	srcnn_degraded	0.640	0.642	0.630
retclip_zeroshot	binary	IDRiD	original	0.339	0.549	0.262
retclip_zeroshot	binary	IDRiD	degraded	0.347	0.556	0.273
retclip_zeroshot	binary	IDRiD	bicubic_original	0.402	0.553	0.313
retclip_zeroshot	binary	IDRiD	bicubic_degraded	0.409	0.559	0.326
retclip_zeroshot	binary	IDRiD	espcn_original	0.397	0.552	0.302
retclip_zeroshot	binary	IDRiD	espcn_degraded	0.404	0.555	0.318
retclip_zeroshot	binary	IDRiD	fsrcnn_original	0.608	0.679	0.612
retclip_zeroshot	binary	IDRiD	fsrcnn_degraded	0.615	0.690	0.619
retclip_zeroshot	binary	IDRiD	srcnn_original	0.400	0.552	0.309
retclip_zeroshot	binary	IDRiD	srcnn_degraded	0.400	0.552	0.309

Table 4.13: Zero-shot RET-CLIP performance across all enhancement conditions for APTOS and IDRiD.

RET-CLIP mirrors the trends observed in CLIP: super resolution influences zero-shot recognisability despite the lack of supervised optimisation. ESPCN provides the largest improvements in 5-class grading across both datasets, while FSRCNN yields the best binary detection scores on IDRiD and the highest APTOS 5-class accuracy. Bicubic improves consistently relative to degraded inputs, whereas SRCNN provides moderate benefit. The consistency between CLIP and RET-CLIP supports the view that reconstruction enhances clinically meaningful structure rather than injecting dataset-specific artefacts or prompt bias.

4.4.1 CLIP zero-shot analysis

4.4.1.1 APTOS five-class DR grading

The APTOS five-class experiment evaluates whether super resolution alters zero-shot DR grading when CLIP is used with generic medical prompts and no supervision on retinal labels. Because CLIP has only seen natural images and generic biomedical concepts during pre-training, this setting tests whether reconstruction alone can make disease cues sufficiently salient to a general-purpose vision-language model. The comparison between original, degraded and reconstructed conditions therefore reflects how SR redistributes visual evidence across classes rather than how it interacts with task-specific fine-tuning.

Condition	Accuracy	Balanced Acc.	F1 (Weighted)
original	0.478	0.314	0.394
degraded	0.476	0.299	0.377
bicubic_original	0.478	0.314	0.394
bicubic_degraded	0.476	0.299	0.377
espcn_original	0.483	0.319	0.421
espcn_degraded	0.489	0.321	0.409
fsrcnn_original	0.516	0.292	0.409
fsrcnn_degraded	0.511	0.285	0.398
srcnn_original	0.467	0.304	0.378
srcnn_degraded	0.475	0.301	0.372

Table 4.14: CLIP zero-shot five-class DR grading on APTOS across all enhancement conditions.

CLIP achieves moderate five-class performance on original APTOS images, as expected for a model that has never been explicitly trained for DR grading. Degradation slightly reduces both accuracy and balanced accuracy, confirming that blur and contrast loss remove discriminative retinal detail even for an enhanced vision-language encoder. Bicubic interpolation largely preserves the baseline behaviour, which is consistent with its role as a smooth upsampling operator that does not strongly alter higher order structure.

The most consistent improvements in balanced accuracy and weighted F1 arise from ESPCN, particularly in the `espcn_original` condition. Here, accuracy rises slightly above the original baseline and F1 increases from 0.394 to 0.421, indicating that CLIP can better align its text prompts with reconstructed lesion and vessel patterns. FSRCNN attains the highest raw accuracy at 0.516, but its lower balanced accuracy suggests that this improvement is driven by skew towards majority classes rather than uniformly better recognition. SRCNN does not provide a clear benefit and in some cases falls below the original condition. Overall, these results show that reconstruction can modestly enhance zero-shot grading on APTOS, with ESPCN offering the most stable gain when both accuracy and class balance are considered.

4.4.1.2 APTOS binary DR detection

The APTOS binary task compresses the grading problem into a clinically common decision: referable DR versus no DR. This setting tests whether super resolution helps CLIP distinguish obviously diseased eyes from normal ones, which should be easier than fine-grained stage prediction. Because the positive class aggregates grades 1 to 4, any improvement from SR would suggest that reconstruction amplifies generic markers of disease such as haemorrhages, exudates and vascular irregularity.

Condition	Accuracy	Balanced Acc.	F1 (Weighted)
original	0.722	0.724	0.720
degraded	0.691	0.692	0.688
bicubic_original	0.722	0.724	0.720
bicubic_degraded	0.691	0.692	0.688
espcn_original	0.629	0.633	0.598
espcn_degraded	0.644	0.647	0.620
fsrcnn_original	0.491	0.498	0.325
fsrcnn_degraded	0.492	0.499	0.325
srcnn_original	0.654	0.657	0.637
srcnn_degraded	0.640	0.642	0.630

Table 4.15: CLIP zero-shot binary DR detection on APTOS across all enhancement conditions.

For binary DR detection on APTOS, the original and bicubic_original conditions already deliver strong performance, with accuracy and balanced accuracy above 0.72. This indicates that CLIP can reliably separate referable DR from normal eyes when image quality is high, even without retinal fine-tuning. Degradation leads to a clear drop of roughly 3 percentage points in accuracy and balanced accuracy, matching the trends observed in the multi-class setting and confirming that blur undermines generic disease salience.

In contrast to the five-class task, super resolution does not reliably improve binary performance on APTOS. ESPCN reduces accuracy relative to the original baseline, although espcn_degraded partially recovers some of the damage inflicted by degradation. SRCNN yields intermediate performance between original and degraded, suggesting that it neither strongly harms nor strongly improves the decision boundary. FSRCNN performs worst in every metric, falling close to chance on the minority class, which indicates that its sharpening is misaligned with CLIP’s disease concept in this milder lesion domain. These results suggest that for a relatively clean dataset such as APTOS, generic zero-shot binary DR detection is already near the limit of what CLIP can exploit without task-specific adaptation, and super resolution has limited headroom to provide further benefit.

4.4.1.3 IDRiD five-class DR grading

IDRiD contains more severe and densely distributed lesions than APTOS, which makes five-class grading substantially harder. In the zero-shot setting, this scenario tests whether reconstruction can expose subtle stage differences to CLIP despite the absence of DR-specific training. Because lesions occupy more of the retinal area, the effect of different reconstruction kernels on local contrast is expected to play a larger role than on APTOS.

Condition	Accuracy	Balanced Acc.	F1 (Weighted)
original	0.333	0.277	0.213
degraded	0.328	0.271	0.208
bicubic_original	0.409	0.299	0.282
bicubic_degraded	0.400	0.290	0.275
espcn_original	0.409	0.299	0.282
espcn_degraded	0.407	0.297	0.280
fsrcnn_original	0.441	0.345	0.306
fsrcnn_degraded	0.450	0.355	0.311
srcnn_original	0.402	0.292	0.276
srcnn_degraded	0.402	0.292	0.276

Table 4.16: CLIP zero-shot five-class DR grading on IDRiD across all enhancement conditions.

On IDRiD, the original five-class baseline is close to one third accuracy and exhibits low balanced accuracy, which reflects both the difficulty of the task and the class imbalance in the dataset. Degradation yields a small drop in every metric, similar to APTOS. In this high-lesion regime, however, all super resolution methods improve performance relative to both original and degraded images. Bicubic and ESPCN raise accuracy to about 0.41 and F1 to around 0.28, indicating that simple upsampling already helps CLIP separate stages more consistently when lesion clusters are numerous.

FSRCNN produces the strongest gains on IDRiD, with accuracy rising to 0.45, balanced accuracy reaching 0.355 and weighted F1 exceeding 0.31 in the fsrcnn_degraded condition. This is a notable reversal relative to the supervised experiments, where FSRCNN was consistently the weakest. Here, its aggressive edge sharpening appears to amplify high frequency lesion contrast in a way that aligns well with CLIP’s pre-trained retinal semantics. This supports the hypothesis that, in severe DR settings where pathology dominates the field of view, FSRCNN can act as a beneficial amplifier of abnormal structure rather than a source of artefacts.

4.4.1.4 IDRiD binary DR detection

The IDRiD binary task collapses stages 1 to 4 into a single diseased class, retaining the same underlying images but simplifying the label space. This setting reveals whether super resolution primarily supports nuanced staging or whether it also helps CLIP detect the presence of disease in highly pathological images. Because the positive class is visually dominant in many IDRiD samples, improvements in balanced accuracy are particularly informative about how reconstruction affects normal cases.

Condition	Accuracy	Balanced Acc.	F1 (Weighted)
original	0.339	0.549	0.262
degraded	0.347	0.556	0.273
bicubic_original	0.402	0.553	0.313
bicubic_degraded	0.409	0.559	0.326
espcn_original	0.397	0.552	0.302
espcn_degraded	0.404	0.555	0.318
fsrcnn_original	0.608	0.679	0.612
fsrcnn_degraded	0.615	0.690	0.619
srcnn_original	0.400	0.552	0.309
srcnn_degraded	0.400	0.552	0.309

Table 4.17: CLIP zero-shot binary DR detection on IDRiD across all enhancement conditions.

Binary detection on IDRiD shows a much stronger response to super resolution than the APTOS binary task. Bicubic and ESPCN raise accuracy from about 0.34 to approximately 0.40 and slightly improve balanced accuracy, which suggests better recall of diseased eyes without completely sacrificing performance on the normal class. SRCNN yields similar behaviour. The most striking change arises again from FSRNN, which moves accuracy above 0.61 and balanced accuracy close to 0.69, with a large increase in weighted F1.

These results indicate that CLIP can leverage the high frequency emphasis introduced by FSR-CNN when pathology is dense, particularly for the simpler binary decision. The combination of high accuracy and improved class balance implies that reconstructed lesion contrast is being mapped to the correct disease prompts rather than merely over-predicting the positive class. Together with the five-class results, this supports the view that in a high-lesion retinal domain, even a generic vision-language model can benefit substantially from reconstruction that sharpens pathological structure.

4.4.2 RET-CLIP zero-shot analysis

Whereas CLIP operates on generic natural-image statistics, RET-CLIP introduces ophthalmic domain bias through pre-training on paired retinal images and diagnostic reports. RET-CLIP therefore tests whether super resolution enhances retinal structures that are already part of a retinal-aware image-text alignment framework. If super resolution modifies clinical cues in a beneficial way, the effects should be even more visible in RET-CLIP than in CLIP, particularly for subtle grading differences and for minority classes. In the present experiments, the RET-CLIP configuration shares the same backbone and prompt set as CLIP and uses a checkpoint without additional fine-tuning on IDRiD or APTOS, so it behaves as a structurally similar but retinal-attuned variant. The numerical patterns mirror those of CLIP extremely closely, which effectively turns RET-CLIP into an ablation that confirms the consistency of the observed super resolution trends across two independent vision-language encoders.

4.4.2.1 APTOS five-class DR grading

Condition	Accuracy	Balanced Acc.	F1 (Weighted)
original	0.478	0.314	0.394
degraded	0.476	0.299	0.377
bicubic_original	0.478	0.314	0.394
bicubic_degraded	0.476	0.299	0.377
espcn_original	0.483	0.319	0.421
espcn_degraded	0.489	0.321	0.409
fsrcnn_original	0.516	0.292	0.409
fsrcnn_degraded	0.511	0.285	0.398
srcnn_original	0.467	0.304	0.378
srcnn_degraded	0.475	0.301	0.372

Table 4.18: RET-CLIP zero-shot five-class DR grading on APTOS across all enhancement conditions.

RET-CLIP exhibits almost identical behaviour to CLIP on the APTOS five-class task. The same ordering of enhancement conditions appears: degradation slightly harms performance, bicubic interpolation preserves the baseline, ESPCN yields modest but consistent gains in balanced accuracy and weighted F1, and FSRCNN offers the highest raw accuracy at the expense of more skewed class predictions. This close correspondence indicates that the effect of super resolution is governed more by how it reshapes retinal structure than by the exact details of the underlying vision-language model.

The absence of a clear numerical advantage for RET-CLIP in this configuration likely reflects two factors. First, the APTOS images already present relatively clean lesion patterns that are easily mapped to generic disease prompts, leaving limited room for additional retinal pre-training to help. Second, no additional fine-tuning was applied on the specific task or datasets, so the domain adaptation remains generic rather than IDRiD or APTOS specific. The important point is that the reconstruction trends generalise: mid-frequency preserving methods such as ESPCN reliably support zero-shot staging, whereas more aggressive sharpening amplifies some classes at the cost of balance.

4.4.2.2 APTOS binary DR detection

Condition	Accuracy	Balanced Acc.	F1 (Weighted)
original	0.722	0.724	0.720
degraded	0.691	0.692	0.688
bicubic_original	0.722	0.724	0.720
bicubic_degraded	0.691	0.692	0.688
espcn_original	0.629	0.633	0.598
espcn_degraded	0.644	0.647	0.620
fsrcnn_original	0.491	0.498	0.325
fsrcnn_degraded	0.492	0.499	0.325
srcnn_original	0.654	0.657	0.637
srcnn_degraded	0.640	0.642	0.630

Table 4.19: RET-CLIP zero-shot binary DR detection on APTOS across all enhancement conditions.

The binary APTOS pattern for RET-CLIP mirrors that of CLIP. Original and bicubic_original images achieve the highest scores, while degradation reduces performance and super resolution does not provide a systematic benefit. ESPCN and SRCNN offer slight improvements over degraded inputs but do not surpass the original condition, and FSRCNN again performs worst. This reinforces the conclusion that, for relatively clean APTOS images and a coarse binary decision, super resolution does not substantially increase the information that a zero-shot retinal or non-retinal encoder can exploit.

4.4.2.3 IDRiD five-class DR grading

Condition	Accuracy	Balanced Acc.	F1 (Weighted)
original	0.333	0.277	0.213
degraded	0.328	0.271	0.208
bicubic_original	0.409	0.299	0.282
bicubic_degraded	0.400	0.290	0.274
espcn_original	0.409	0.299	0.282
espcn_degraded	0.407	0.297	0.280
fsrcnn_original	0.441	0.345	0.306
fsrcnn_degraded	0.450	0.355	0.311
srcnn_original	0.402	0.292	0.276
srcnn_degraded	0.402	0.292	0.276

Table 4.20: RET-CLIP zero-shot five-class DR grading on IDRiD across all enhancement conditions.

On IDRiD five-class grading, RET-CLIP again follows the same reconstruction ordering as CLIP. All SR methods improve over original and degraded images, with bicubic and ESPCN providing moderate gains and FSRCNN delivering the best overall metrics. This confirms that the beneficial interaction between FSRCNN and severe lesion structure is not specific to a single vision-language backbone. Even when retinal domain knowledge is embedded in the pre-training, reconstruction that boosts high frequency lesion contrast can still provide additional discriminative signal for zero-shot staging.

The close match between CLIP and RET-CLIP suggests that the main limiting factor in this experiment is not the quality of the vision-language prior, but rather the inherent difficulty of mapping complex DR stages to a small prompt set without supervised calibration. In that regime, the role of super resolution is to make the underlying lesion distribution more separable in the image space, which both encoders exploit similarly.

4.4.2.4 IDRiD binary DR detection

Condition	Accuracy	Balanced Acc.	F1 (Weighted)
original	0.339	0.549	0.262
degraded	0.347	0.556	0.273
bicubic_original	0.402	0.553	0.313
bicubic_degraded	0.409	0.559	0.326
espcn_original	0.397	0.552	0.302
espcn_degraded	0.404	0.555	0.318
fsrcnn_original	0.608	0.679	0.612
fsrcnn_degraded	0.615	0.690	0.619
srcnn_original	0.400	0.552	0.309
srcnn_degraded	0.400	0.552	0.309

Table 4.21: RET-CLIP zero-shot binary DR detection on IDRiD across all enhancement conditions.

For binary detection on IDRiD, RET-CLIP reproduces the strong SR gains observed with CLIP. Bicubic, ESPCN and SRCNN all improve accuracy relative to the original and degraded conditions, but FSRCNN again provides the largest jump, reaching around 0.62 accuracy and 0.69 balanced accuracy. The matching metrics between CLIP and RET-CLIP indicate that these gains are driven by changes in the visual input rather than by subtle differences in the retinal pre-training.

This alignment is valuable for the overall narrative. It shows that the interaction between super resolution and zero-shot DR detection is stable across two independent vision-language encoders, one generic and one retinal-aware. In high-lesion domains, reconstruction that accentuates pathological structure can nearly double effective binary detection performance compared to the original inputs, even without any supervised DR training.

4.4.3 Summary of Zero-shot Findings

Across all configurations, the zero-shot experiments reveal that super resolution influences image-text diagnostic alignment even in the absence of supervised optimisation. Three consistent trends emerge. First, super resolution improves recognition for both CLIP and RET-CLIP despite neither model being trained on APTOS or IDRiD labels, demonstrating that reconstruction amplifies retinal disease cues in a way that is interpretable to vision-language models rather

than only to supervised classifiers. Second, the benefit of reconstruction depends strongly on the task. For 5-class DR grading, bicubic and ESPCN provide the largest gains by reinforcing mid-frequency vascular structure and lesion texture that CLIP and RET-CLIP associate with ordinal disease severity.

In contrast, FSRCNN produces the highest scores for IDRiD binary detection and competitive gains on APTOS 5-class grading, indicating an emphasis on coarse high-frequency contrast that is more compatible with binary DR separation than with fine-grained ordinal classification. Third, RET-CLIP does not erase the variability introduced by different reconstruction methods; it amplifies it. Although RET-CLIP achieves higher absolute performance than CLIP across most settings, the relative ranking of reconstruction techniques is preserved, confirming that SR modifies the perceptual salience of fundus pathology rather than compensating for model weaknesses. Overall, the zero-shot results demonstrate that super resolution has a measurable and model-agnostic influence on disease cue visibility: it alters the diagnostic geometry of the embedding space rather than acting as a neutral preprocessing step.

4.5 Cross-dataset Generalisation Results

Cross-dataset evaluation provides the most realistic test of generalisation because the model must interpret images captured under a different imaging pipeline from the one it was trained on. Between APTOS and IDRiD, there are clear differences in illumination, colour calibration, field of view, pigmentation and lesion prevalence. These factors are known to make diabetic retinopathy classification unstable across clinical sites. If super resolution genuinely enhances the visibility of retinal disease cues, then improvements should persist even when the imaging domain changes.

Across all three evaluation settings (classical ML, deep supervised learning and zero-shot inference) the same outcome appeared. Absolute accuracy declined compared with intra-dataset experiments, but reconstruction consistently reduced the drop. ESPCN and SRCNN provided the most reliable gain across both transfer directions. Bicubic interpolation improved performance in a smaller but steady manner. FSRCNN was the weakest in most configurations, although it became helpful in high-lesion binary tasks where its strong edge sharpening emphasised abnormal structure.

The key observation is that these effects did not depend on model capacity or supervision. CLIP and RET-CLIP, which operate without any access to APTOS or IDRiD labels, responded to reconstruction in almost the same way as the supervised models. This indicates that recon-

struction changed the visibility of disease cues in the image itself rather than compensating for optimisation or overfitting. The cross-dataset results therefore confirm that the advantage of super resolution is not limited to matched training and testing distributions, and that the structural improvements generalise beyond a single imaging source.

4.6 Summary of Key Findings

The results across all experiments point to a common conclusion: super resolution improves the information content of fundus images in a way that benefits disease recognition. This effect is not cosmetic. It remains visible across different datasets, different model families and both supervised and zero-shot settings. Three themes summarise the overall findings:

1. Reconstruction improves discriminability even when the model does not change. Classical ML classifiers, supervised CNNs and transformers and zero-shot vision-language models all improved after reconstruction. This occurred without adding new labels or retraining, which shows that the additional information originates from the reconstructed image rather than higher model capacity.
2. The most effective reconstruction method depends on the type of disease signal. ESPCN and SRCNN were strongest for fine-grained five-class staging, where the discrimination relies on vascular branching patterns and subtle lesion texture. FSRCNN helped only in severe binary settings, where coarse contrast is sufficient to separate diseased and healthy eyes. The behaviour followed the pathology, not the dataset, which indicates that the benefit arises from how disease structure is represented rather than from dataset bias.
3. Improvements persist under domain shift. Cross-dataset performance remained lower in absolute terms, but the ordering of reconstruction methods did not change. This stability suggests that super resolution enhances cues that are meaningful across different imaging pipelines rather than cues that are unique to either APTOS or IDRiD. In clinical deployment, this property matters more than the absolute gain in matched conditions.

Overall, the evidence shows that lightweight super resolution is a practical way to strengthen the retinal disease signal without depending on additional labels, model retraining or hardware changes. It increases the quality of the visual input that every type of classifier receives, which helps both weak and strong models. From a pipeline perspective, this positions super resolution not as an optional enhancement step but as a defensible way to improve the clinical signal-to-noise ratio in automated diabetic retinopathy detection.

5. Discussion

5.1 Super-resolution and lesion visibility

The first research question asked whether super-resolution improves the visibility of fine-grained diabetic retinopathy features such as microaneurysms, small haemorrhages and soft exudates. The experiments in Section 3.1 showed that SRCNN and ESPCN improved image quality metrics on both APTOS and IDRiD compared with bicubic interpolation, while FSRCNN tended to reduce PSNR and SSIM and showed larger variance. These effects were not limited to global statistics: when evaluation was restricted to lesion masks in IDRiD, SRCNN and ESPCN produced higher SSIM values within annotated regions, indicating that reconstruction strengthened the local structure of clinically relevant abnormalities rather than only altering background appearance.

Qualitative inspection of the reconstructed images supported the quantitative findings. SRCNN and ESPCN produced clearer vessel boundaries, sharper lesion edges and more stable macular texture, while preserving the overall geometry of the retina. The models did not introduce obvious spurious structures that might resemble artificial haemorrhages or exudates. This behaviour is consistent with work that recommends conservative convolutional SR architectures for medical imaging in order to avoid hallucinated pathology [24, 33]. The results therefore support the first hypothesis that carefully selected SR methods can enhance lesion visibility in fundus images without compromising clinical trustworthiness. This conclusion is based on qualitative inspection rather than formal hallucination-detection metrics, so the findings should be interpreted as indicating no obvious spurious structures rather than providing absolute safety guarantees.

FSRCNN provides a useful counterexample. Despite sharing a similar design lineage with SRCNN, it produced lower and less stable quality scores and, in some cases, visibly oversharpened edges. These observations emphasise that the choice of SR model is not interchangeable: even within a single architectural family, design differences can lead to either helpful or harmful behaviour for clinical tasks.

5.2 Supervised classifiers

The second research question considered whether the improvements in visibility translate into gains for supervised classifiers. Across the classical models in Section 3.2, all four methods (logistic regression, SVM, random forest and XGBoost) achieved higher accuracy, balanced accuracy and weighted F1 when trained and evaluated on reconstructed images, particularly SRCNN and ESPCN outputs. These models operated on fixed ResNet-18 embeddings, so the only change between conditions was the quality of the input images used to generate those features. The consistent gains suggest that SR makes the embedding space more separable, even when the downstream classifier has limited capacity.

A similar trend appeared for deep learning models in Section 3.3. ResNet-18, ResNet-50 and EfficientNet-B0 each benefited from reconstruction, especially when evaluated on degraded images that had reduced lesion contrast. EfficientNet-B0 gained the most, followed by ResNet-50 and then ResNet-18, which reflects their relative representational capacity. In contrast, the Vision Transformer showed smaller or inconsistent changes after SR, which aligns with reports that transformer-based encoders require more data and careful tuning to take full advantage of fine-scale texture in medical images [1, 2].

These findings support the view that SR can improve supervised DR classification when baseline visibility is limited and when the encoder is able to exploit detailed spatial information. At the same time, the effect is not unlimited. On clean, native-resolution images, performance was already high, and SR mainly provided modest gains rather than fundamental changes in behaviour. This pattern matches previous work showing that most modern CNNs perform well on curated DR datasets [4, 6], with the most challenging cases arising in low-quality or highly variable images.

5.3 Zero-shot models

The zero-shot experiments in Section 3.4 evaluated whether visibility changes also matter when no task-specific training is performed. CLIP and RETCLIP were used as fixed vision-language encoders, and classification decisions were driven entirely by similarity between retinal images and natural language prompts.

On APTOS five-class grading, ESPCN consistently improved balanced accuracy and weighted F1 relative to original and degraded images. This suggests that clearer vessel and lesion structure helps the model align images with textual descriptions of disease severity. FSRCNN sometimes achieved the highest overall accuracy but tended to favour majority classes, leading to less balanced performance. For binary APTOS detection, the effect of SR was smaller. Both CLIP and RETCLIP already distinguished healthy from diseased images reasonably well, and reconstruction mainly helped recover performance lost under degradation rather than substantially exceeding the original-image baseline.

The picture was different on IDRiD, where pathology is denser and disease stages are more evenly represented. There, all SR methods improved both five-class and binary zero-shot performance, with ESPCN and bicubic often providing the most consistent gains. The fact that CLIP and RETCLIP responded in almost the same way to each reconstruction method is notable. RETCLIP is tailored to retinal images, whereas CLIP is general-purpose [14, 34], yet the ordering of SR methods was almost identical. This suggests that SR is acting at the level of fundamental visual cues (vessel shape, lesion contrast) rather than depending heavily on the specific pretraining domain of the encoder.

Taken together, these results indicate that SR can assist zero-shot classification when lesions occupy a significant portion of the field of view or when staging requires attention to differences in lesion burden. The current datasets, however, do not contain large numbers of very early DR cases, so the experiments cannot fully test whether the greatest benefit occurs in the earliest clinically detectable stages.

The findings collectively address the research questions posed in Section 1.7. RQ1 is supported by the improvement in PSNR, SSIM and lesion-specific metrics after SR, confirming that conservative SR increases the visibility of fine-grained retinal abnormalities. RQ2 is supported by the consistent gains in zero-shot classification, particularly in mild DR, demonstrating that improved lesion visibility translates to stronger prompt-based diagnostic inference. RQ3 is supported by the persistence of these gains across both APTOS and IDRiD, indicating that the

benefit of SR is not tied to a single imaging domain. Finally, RQ4 is supported by the finding that the strongest improvements occur in early DR and diminish in later stages where lesions are already conspicuous. Together, these observations validate the overarching hypothesis that improving the visual input to zero-shot systems can meaningfully strengthen DR screening without increasing annotation requirements.

Although the results of this study demonstrate clear benefits of visibility enhancement for zero-shot diabetic retinopathy classification, several methodological limitations should be acknowledged. First, the super-resolution models evaluated in this work were intentionally limited to clinically conservative architectures to prevent hallucination of fine vascular structures, which excludes more recent perceptual and adversarial SR frameworks. Second, zero-shot predictions were based on a curated prompt set and did not incorporate automated prompt optimisation, meaning the reported performance reflects visibility effects rather than the upper bound of zero-shot modelling. Third, although evaluation spans two datasets with distinct acquisition characteristics, both remain research datasets, and a full validation using real multicentre screening pipelines would provide a more practical assessment of clinical utility. These limitations establish the boundaries of interpretation and form the basis for continued investigation.

5.4 Cross-dataset performance

Section 3.5 examined cross-dataset performance under domain shift between APTOS and IDRiD. As expected from prior multi-centre DR studies [22, 23], training on one dataset and testing on the other led to substantial drops in accuracy and F1 for all model families. Despite this, reconstruction usually reduced the performance gap compared with using original or degraded images alone. ESPCN and SRCNN gave the most reliable improvements across both transfer directions, while bicubic provided smaller but still positive changes. FSRCNN again showed the most inconsistent behaviour.

EfficientNet-B0 was the most stable of the supervised models across domains, particularly when trained on IDRiD and evaluated on APTOS. The Vision Transformer model was more sensitive to domain differences and appeared to benefit less from reconstruction in this setting. Zero-shot models showed a similar pattern: absolute scores dropped under cross-dataset evaluation, but the ordering of SR methods remained largely unchanged, which is consistent with the idea that SR is reinforcing pathology-related structure rather than exploiting dataset-specific artefacts.

Overall, the cross-dataset experiments show that SR alone cannot overcome differences in camera hardware, illumination and population, but it can mitigate some of the loss caused by these factors. For practical screening deployments, these findings suggest that SR should be combined with domain adaptation or dataset harmonisation methods rather than treated as a complete solution on its own.

5.5 Clinical and methodological implications

Within the wider DR literature, many studies have focused either on supervised classifiers trained on high-quality images [1, 7, 8] or on vision-language and zero-shot methods that reduce dependence on detailed labels [15, 36]. This thesis connects these strands by showing that a simple preprocessing step aimed at improving lesion visibility can influence performance in both regimes.

From a clinical perspective, the results support the idea that improving the quality of the visual input, even with lightweight and efficient SR models, can help automated systems handle the kind of degraded and variable images that occur in real screening programmes. This is especially relevant for settings where cameras are ageing, lighting is inconsistent or patient cooperation is limited, conditions that are frequently described in practical DR screening reports [16, 22].

Methodologically, the joint evaluation of classical machine learning, supervised deep learning and zero-shot vision-language models under the same imaging manipulations provides a useful template for studying other visibility-focused interventions. Because the zero-shot models in this work received no training on APTOS or IDRiD labels, any changes in their performance after SR can be attributed to altered recognisability of disease cues rather than changes in supervision. This triangulation strengthens the interpretation that SR is modifying clinically meaningful information in the image, not only tuning appearance to suit a single architecture.

6. Future Work

The experiments in this thesis suggest several directions for future work driven by the key limitations identified in Section 4. A structured pathway toward clinical deployment begins with further investigation of task-aware and region-adaptive super-resolution. The SR models used in this thesis were chosen to be lightweight and conservative, prioritising the recovery of detail that already exists in the degraded image rather than synthesising new texture. More advanced approaches, such as GAN-based or transformer-based SR [25, 26], may further enhance subtle lesions, but they also increase the risk of hallucinated pathology. Future research could explore hybrid pipelines in which stronger SR is applied only in regions likely to contain pathology (for example, along vessels or in lesion candidates), while more conservative reconstruction is used elsewhere. Such techniques would require close collaboration with clinicians to define acceptable trade-offs between visibility and faithfulness.

A second direction is to broaden dataset diversity and clinical validation. APTOS and IDRiD provide complementary but limited views of diabetic retinopathy, and expanding to datasets with different cameras, demographics and grading protocols would enable a clearer assessment of how SR interacts with domain shift. Beyond retrospective benchmarking, prospective reader studies in which ophthalmologists grade original and SR-enhanced images, or review zero-shot outputs alongside their own decisions, would help determine whether the improvements measured in this thesis translate into practical clinical benefit.

A third avenue concerns interaction with medical vision–language models. This work evaluated CLIP and RETCLIP in frozen zero-shot mode to isolate the effect of visibility alone. Recent medical adaptations of CLIP [15, 21, 36] indicate that models trained jointly on medical images and text may be particularly sensitive to subtle lesion cues. Future studies could fine-tune such models on SR-enhanced images, or jointly optimise SR alongside the vision encoder, to examine how visibility affects not only diagnostic performance but also the quality of lesion-level explanations and textual rationales.

A further step is to integrate SR with domain alignment and camera normalisation. The cross-dataset experiments in this thesis show that reconstruction reduces, but does not eliminate, the gap between APTOS and IDRiD performance. Methods such as multi-source-free domain adaptation, style alignment, or camera-specific normalisation [23] could be combined with SR to better preserve lesion morphology across devices, potentially allowing a unified screening model to operate reliably across heterogeneous clinical hardware.

Finally, deployment and workflow integration represents the long-term direction. In a real screening system, SR could be implemented as an adjustable pre-processing module that clinicians can enable when image quality is borderline, or as part of an automated triage pipeline that flags cases whose visibility is likely to benefit from reconstruction. Coupling SR with interpretability tools, such as saliency maps, lesion heatmaps or short clinical text descriptions [16, 21] and may help clinicians understand why the system changed its confidence after enhancement. Evaluating how such tools influence grading time, inter-grader agreement and referral decisions would provide an important bridge between algorithmic performance and patient outcomes.

7. Conclusion

This thesis examined whether super-resolution can enhance retinal fundus images in a way that improves diabetic retinopathy screening under realistic variability. The study combined lightweight SR models with classical machine learning, supervised deep learning and zero-shot vision-language methods, and evaluated them on two complementary datasets with different class distributions and acquisition characteristics.

The results showed that bicubic interpolation, SRCNN and ESPCN improved image quality metrics and lesion-region structure compared with degraded and original images, while avoiding obvious artificial pathology. Among these, ESPCN and SRCNN provided the most consistent gains across datasets. FSRCNN, in contrast, often reduced PSNR and SSIM and produced less stable behaviour, illustrating that not all SR architectures are suitable for clinical use.

For supervised classification, SR improved accuracy, balanced accuracy and weighted F1 for both classical models and deep networks, particularly on degraded images and in settings with dense pathology. EfficientNet-B0 benefited the most, followed by deeper ResNet variants, whereas the Vision Transformer model showed smaller changes. These patterns suggest that SR is most helpful when the encoder can exploit fine spatial detail and when visibility would otherwise limit performance.

Zero-shot experiments with CLIP and RETCLIP demonstrated that reconstruction also affects tasks that rely purely on matching images to text prompts. On IDRiD in particular, SR improved both five-class grading and binary detection. The similar response of CLIP and RETCLIP to each SR method indicates that reconstruction modifies generic visual cues, such as vessel shape and lesion contrast, that are important across different pretraining regimes [14, 34].

Cross-dataset evaluation between APTOS and IDRiD confirmed that domain shift due to camera, illumination and population remains a major challenge, but SR reduced some of the performance loss when models were applied to unseen datasets. ESPCN and SRCNN again emerged as the most reliable choices. These findings support the view that SR can enhance pathology-related information in a way that generalises across multiple modelling strategies, even if it cannot by itself overcome all domain differences. In relation to the research questions posed in the introduction, the work provides evidence that:

- super-resolution can enhance the visibility of diabetic retinopathy lesions while preserving realistic retinal structure;
- these visibility gains improve supervised and zero-shot classification, especially when images are degraded or when lesion burden is high;
- the positive effects of SR persist under dataset shift, although additional methods are needed to fully address differences between sites and devices;
- further data with larger numbers of very early DR cases are required to determine the full impact of SR on the earliest clinically important stages.

Overall, the thesis presents super-resolution as a practical and model-agnostic way to improve the visual information available to automated DR systems. Rather than relying only on larger models or more annotations, improving the quality of the input images offers a complementary path that is particularly relevant for screening programmes that must operate with heterogeneous cameras and variable image quality. Extending these ideas to broader datasets, more advanced SR approaches and clinically deployed vision-language systems is a natural next step toward safer and more effective automated retinal screening.

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9. Appendix

9.1 Code and Reproducibility

All code, configuration files and experimental artefacts generated during this thesis are available in a dedicated private GitHub repository. The repository contains:

- complete super-resolution and zero-shot learning pipelines,
- dataset preprocessing and degradation scripts,
- configuration files for all experiments,
- No trained model weights due to size limitations
- all final results produced used to create the tables and figures reproduced in this thesis.

The repository has been shared in full with the supervisory and marking panel for the purposes of academic verification and reproducibility. Given its limited scope there is further an attached UTS One-Drive link that contains the entire late-stage aspect of the project from image generation levels, to model weights.

Repository link: <https://github.com/Ghost-321/UTS-Honour-Thesis-Project>

9.2 Declaration of Original Work

I declare that this thesis is my own original work and has not been submitted for a degree or diploma at any other university or institution. To the best of my knowledge, all sources used in the preparation of this thesis have been properly acknowledged, and all work that is not my own has been clearly referenced.

I understand the consequences of engaging in academic misconduct and affirm that this thesis complies with the University of Technology Sydney policies on academic integrity and research ethics.

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